

Lanthanides or Rare Earths (4f-Block Elements)

DEFINITION

The elements in which the additional electron enters $(n-2)f$ orbitals are called inner transition elements. The valence-shell electronic configuration of these elements can be represented as: $(n-2)f^{0,2\cdots 14} (n-1)d^{0,1,2} ns^2$. These are also called *f-block elements* because the extra electron goes to *f*-orbitals which belong to $(n-2)$ th main shell. *f*-block elements have been classified as 4*f* and 5*f* block elements corresponding to the filling of 4*f* and 5*f* orbitals of $(n-2)$ th main shell.

The complete and valence-shell electronic configurations of 4*f* and 5*f*-block elements can be represented as: (*n* represents the outermost shell).

Block elements	Complete configuration	Valence-shell configuration
4 <i>f</i> -block elements	2, 8, 18, 4 <i>s</i> ² <i>p</i> ⁶ <i>d</i> ¹⁰ <i>f</i> ^{0, 2...14} , 5 <i>s</i> ² <i>p</i> ⁶ <i>d</i> ^{0, 1} , 6 <i>s</i> ²	4 <i>f</i> ^{0, 2...14} , 5 <i>d</i> ^{0, 1} , 6 <i>s</i> ² (<i>n</i> =6)
5 <i>f</i> -block elements	2, 8, 18, 32, 5 <i>s</i> ² <i>p</i> ⁶ <i>d</i> ¹⁰ <i>f</i> ^{0, 2...14} , 6 <i>s</i> ² <i>p</i> ⁶ <i>d</i> ^{0, 1, 2} , 7 <i>s</i> ²	5 <i>f</i> ^{0, 2...14} , 6 <i>d</i> ^{0, 1, 2} , 7 <i>s</i> ² (<i>n</i> =7)

4*f*-block elements are also called *lanthanides*, *lanthanoids* or *rare-earths*. The first two names are given because of their strong resemblance to lanthanum. The name rare earth was given to them because they were originally extracted from oxides for which ancient name was *earth* and which were considered to be rare. The term *rare-earths* is avoided now because many of these elements are no

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longer rare but are abundant. Thus strictly according to this definition of lanthanides only thirteen elements from Ce₅₈ (4*f*²5*d*⁰6*s*²) to Yb₇₀ (4*f*¹⁴5*d*⁰6*s*²) should be the members of lanthanide series. However, all the fifteen elements from La₅₇ (5*d*¹6*s*²) to Lu₇₁ (4*f*¹⁴5*d*¹6*s*²) are considered as the members of lanthanide series, since all the fifteen elements have similar physical and chemical properties. In fact the name *lanthanides* has been derived from lanthanum which is the prototype of lanthanides.

5*f*-block elements are called actinides or actinones.

Lanthanides constitute the *First Inner Transition Series* while actinides constitute the *Second Inner Transition Series*.

POSITION OF LANTHANIDES IN THE PERIODIC TABLE

All the fifteen lanthanides have atomic weights between those of barium (atomic number, *Z*=56) and hafnium (*Z*=72) and, therefore, must be placed between these two elements as was also proved by Moseley.

Barium has exactly the same outer electronic configuration as calcium and strontium and resembles them very closely. In a similar way hafnium is similar to zirconium (*Z*=40). Therefore Ba must be placed below Sr (Group II B) and Hf below Zr (Group IV B), thus leaving only one place between them, which lies exactly below Y (*Z*=39, Group III B).

Since all the fifteen lanthanides resemble one another in many respects, these must be placed in the same group. These elements also resemble Y₃₉ because of the following additional similarities: (a) Owing to lanthanide contraction (to be discussed subsequently), the ionic radius of Y³⁺ ion is almost similar to that of Er³⁺ (*r*_{Y³⁺} = 0.93 Å, *r*_{Er³⁺} = 0.96 Å). (b) Y generally occurs in nature associated with the ores of heavier lanthanides and resembles Tb (III) and Dy (III) in its compounds. It, therefore, became necessary to accommodate all the fifteen lanthanides together at one place. This has been done by placing the first element viz. La₅₇ below Y₃₉ and remaining fourteen elements (viz. Ce₅₈ to Lu₇₁) have been placed separately in the lower part of the periodic table.

OCCURRENCE OF LANTHANIDES

The lanthanides are not so scarce in the earth's crust as the name rare-earths implies. Potentially their sources are unlimited. Each lanthanide mineral contains all lanthanides, although some in traces only, excepting Pm which is an unstable radioactive element.

Classification of Rare-Earth Minerals. The rare-earth minerals, like the earths themselves, fall into two distinct classes.

(1) *Cerium Group Minerals.* These are also called *cerite-earth minerals* or simply *cerium earths* and are rich in cerite-earths. This group of minerals contains largely elements of atomic numbers from 57 to 63 (i.e. La₅₇ — Eu₆₃), the light lanthanide elements. These are thus *light rare earths*.

(2) *Yttrium Group Minerals.* These are also called *yttrium-earth minerals* or simply *yttrium earths* and are rich in gadolinite

Table 9-1. Important Minerals of Lanthanides.

Minerals	Composition	Location of significant deposits
(i) Monazite Sand—Mixture of orthophosphates of Ce-earths, (Ce) PO_4	(1) Cerium group minerals 50–70% Ce-earths (<i>i.e.</i> elements of at. no. 57 to 62 calculated as oxides) 1–4% Y-earths (<i>i.e.</i> elements of at. no. 63 to 71 calculated as oxides) 5–10% TbO_2 1–2% SiO_2 22–30% P_2O_5 Traces of U 65–70% Ce-earths < 1% Y-earths Traces of Thorium 51–72% Ce-earths 7–6% Y-earths Traces of Th, U, Zr	Occurs in the sand beaches of Travancore (India) Brazil South Africa U.S.A. Sweden, California, New Mexico
(ii) Bastnaesite-cerium earth fluoro-carbonate, (Ce) FCO_3		
(iii) $Ca_2(Fe, Al)_2Si_2O_7 \cdot xH_2O$ —A hydrated silicate of the composition, $(Ce)_x M II H_3 Si_3 O_{11} (M = Ca, Fe)$		

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(i) Gadolinite or Ytterbite—A ytterbium-earths, iron and beryllium silicate, $(Fe, Be)_2 (Y_2) Si_2 O_7$	(2) Yttrium group minerals 35–48% Y-earths (Calculated as oxides) 2–17% Ce-earths Upto 11.6% BeO Traces of ThO_2 54–65% Y-earths ~ 0.1% Ce-earths	Sweden, Norway USA (Texas and Colorado)
(ii) Xenotime—An orthophosphate of Y-earths (analogous to Monazite), $(Y) PO_4$	Upto 3% ThO_2 , upto 3.5% U_2O_5 2–3% ZrO_2 13–35% Y-earths (Calculated as Oxides) 2–8% Ce-earths (Calculated as Oxides) 20–23% ThO_2 , 25–35% (Nb, Ta) O_5	Norway Brazil Australia, Idaho (U.S.A.)
(iii) Xenotime—Mixture of titanates, niobates and tantalates of Y-earths, $(Y) (Nb, Ta, Ti) O_6 \cdot xH_2O$		

earths. This group contains largely yttrium and lanthanides of atomic numbers from 64 to 71 (Gd_{64} — Lu_{71}), the heavier lanthanides. These are thus heavy rare-earths.

Some of the important minerals of these two groups, their composition and the places where their most important deposits are found are given in Table 9-1. In this table the symbols Ce and Y stand for any lanthanide atom of cerium group and yttrium group minerals respectively. Both Euxenite and Gadolinite are quite abundant and are the future sources of yttrium earths.

Besides rare-earth minerals proper, there are many others, particularly the phosphates, arsenates, molybdates and tungstates of Ca, Sr, Fe and Pb which contain small amounts of rare-earths. In some of these Sm, Gd and Eu are present in much larger quantities than in rare-earth minerals.

EXTRACTION OF LANTHANIDES FROM MONAZITE MINERAL

Lanthanides are mainly extracted from monazite mineral which is chiefly worked for thorium and the lanthanides are obtained as by-products. The operation consists of the following three steps:

(I) Concentration of the mineral.

The concentration of monazite sand is usually begun with gravity separation on Wilfley tables, such as those which are used for gold. Since monazite sand is much more heavier than quartz (sp. gravity of monazite=5.5, sp. gravity of quartz=2.6), the monazite sand gets caught up on the riffles and the quartz sand and other gangue material is washed off. The heavier material which contains monazite sand and other material sands (ilemenite and garnet) is dried and then passed through magnetic separator. The monazite, being much less magnetic than the other material sands, gets thereby separated from the rest. The refined monazite so obtained has an average composition $ThO_2=7.5$, $Ce_2O_3=30\%$, other rare earths=32%, $P_2O_5=29\%$ and $SiO_2=1.5\%$.

(II) Cracking of the mineral.

The concentrated monazite obtained as above is subjected to chemical treatment which is technically known as "processing", "opening up" or "cracking" of the mineral.

The chemical treatment of concentrated monazite is done either by conc. H_2SO_4 or by conc. $NaOH$ solution.

(i) Cracking of the mineral by H_2SO_4 (and removal of thorium and cerium). The finely powdered (ca. 100 mesh) and concentrated monazite mineral is heated with 93% H_2SO_4 at $210^\circ C$ in vessels constructed of cast iron, tantalum iron or high silicon iron and provided with mechanical stirrers. Gas or oil-fired furnace is used. The reaction is exothermic. At the end of about four hours, the viscous paste consisting of lanthanide sulphates and thorium sulphate is leached with water for about 12 to 15 hours when all the

(2) This paste

3482 of sand
contain max amount of Lanthanides
sulphates of Ln^{3+} , Th^{4+} and ZrO_2^{2+} (traces) go into solution leaving behind insoluble SiO_2 , unreacted monazite, TiO_2 , $ZrSiO_4$. The latter residue is crushed and returned for extraction. The solution so obtained [the solution becomes acidic due to the formation of H_2PO_4 in the reaction of P_2O_5 with $H_2O: P_2O_5 + 3H_2O \rightarrow 2H_3PO_4 \rightarrow 2H^+ + 2H_2PO_4^-$] which contains the sulphates of Ln^{3+} , Th^{4+} and ZrO_2^{2+} (traces) is treated with sodium pyrophosphate, $Na_2P_2O_7$, to precipitate much of thorium as $Th(P_2O_7)_2$. The filtrate obtained is treated with oxalic acid ($H_2C_2O_4$) to get the precipitate mixture containing $Ln_2(C_2O_4)_3$ and traces of $Th(C_2O_4)_2$ and $(ZrO)_2(C_2O_4)_3$. This mixture is boiled with $(NH_4)_2C_2O_4$ solution whereby traces of $Th(C_2O_4)_2$ and $(ZrO)_2(C_2O_4)_3$ remain in solution and $Ln_2(C_2O_4)_3$ oxalates are left behind as residue which is converted into the corresponding sulphates by igniting carefully with conc. H_2SO_4 . Na_2SO_4 is added to the clear solution of sulphates of Ln^{3+} so obtained so that the light lanthanides (La_{57} to Eu_{63}) precipitate out as double sulphates and the heavy lanthanides (Gd_{64} to Lu_{71}) remain in solution as simple sulphates. Hot $NaOH$ is added to the precipitated mixture containing double sulphates of light lanthanides to form a mixture of hydrated oxides. This mixture is dried in air at $100^\circ C$ whereby we get the mixture of oxides whose approximate composition is $La_2O_3=17\%$, $CeO_2=5\%$, $Pr_2O_3=8\%$, $Nd_2O_3=20\%$, $Sm_2O_3=5\%$ and traces of Eu_2O_3 . This mixture of oxides is treated with dil. HNO_3 when all the light lanthanide oxides (except CeO_2) get dissolved as nitrates and CeO_2 is obtained as crude residue by filtration. The crude CeO_2 is dissolved in 85% HNO_3 to convert it into crude $Ce(NO_3)_3$ which is converted into red basic nitrate, $Ce(OH)(NO_3)_2 \cdot 3H_2O$ by the addition of dil. H_2SO_4 . The solution of nitrates of La^{3+} , Pr^{3+} , Nd^{3+} , Sm^{3+} and Eu^{3+} obtained as above after the removal of CeO_2 is separated into individual components by different methods discussed below.

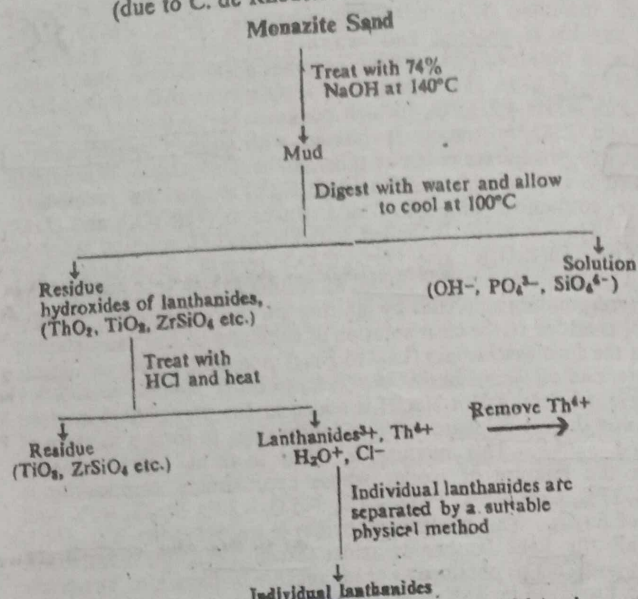
The solution containing the sulphates of heavy lanthanides can also be used to separate the individual components by the same methods as given below.

Thorium can also be removed by other methods like (i) selective precipitation method which consists of adding H_2F_2 to the acidic solution containing Ln^{3+} and Th^{4+} ions when Th^{4+} is precipitated as ThF_4 and that the fluorides of Ln^{3+} ions (LnF_3) go into the solution. Other selective precipitation methods include precipitation of Th^{4+} as ThO_2 with $NaOH$ at pH=5.8 to 6.0. (ii) solvent extraction method which is now widely used to get thorium to be used for nuclear purposes. Organic solvent tri-n-butyl phosphate in xylene is used which preferentially extracts thorium.

Cerium can also be removed by various other methods some of which are: (i) selected crystallisation of the double ammonium nitrate, $Ce(NO_3)_3 \cdot 2NH_4NO_3$ from nitric acid solutions, but the recovery is not quantitative. (ii) selective oxidation, anodically or chemically, with bromate or permanganate results in the precipitation of Ce^{4+} from the buffered solution. (iii) Selective hydrolysis: the concentrated mixture of Ce^{4+} and nitrates of lanthanides (+3) or sulphates is diluted and boiled when Ce^{4+} separates out. (iv) Solvent extraction of Ce^{4+} with tri-n-butyl phosphate is similar to that of Th^{4+} . The Ce^{4+} is reduced to Ce^{3+} and extracted back into aqueous phase by treating the organic phase with aqueous sodium nitrite.

by base (ii) Cracking of the mineral by $NaOH$. This is shown below in Flow-sheet 1.

Flow-sheet 1: Cracking of the monazite mineral by Sodium hydroxide (due to C. de Rhoden and M. Peltier, 1957)



Sulphuric acid vs. caustic soda cracking. Although a greater variety of monazite sands can be processed by H_2SO_4 treatment than by the NaOH method, higher yields and cleaner separations are the main advantages of the latter process. However, the H_2SO_4 treatment method has the advantage over the NaOH process of being slightly cheaper to operate. In the NaOH cracking of monazite, concentration of the rare-earths is achieved at relatively early stage of the process.

(III) Separation of individual rare-earth elements.

The long story of the methods used for the separation of the individual rare-earths may broadly be divided into two classes:

(1) Old-classical methods.

The following classical methods were not only used by the early workers in the past, but still remain as very important methods for economical production of rare-earths on commercial scale.

(i) *Fractional Crystallisation.* This procedure depends on slight differences in the solubility of various salts. The chief salts (either simple or complex) utilised in this procedure are given in Table 9.2. The order of separation is indicated in the last column, the least soluble being placed first.

The mixed solution of rare-earth salts is evaporated to such an extent that, if it is allowed to cool, it would deposit about one-half of the dissolved salts as crystals.

The crystals obtained as a result of the first evaporation are re-dissolved in just sufficient amount of water and the solution is evaporated as before to such an extent that on cooling, it would deposit about one-half of the dissolved salts as crystals. Similarly, the liquor from the first crystallisation is also evaporated further so that about one-half of the dissolved salt may be obtained as crystals.

The crystals and mother liquors obtained from the second set of operations are separated from each other, and the mother liquor from the first crystallisation is combined with the crystals from the second evaporation of the original mother liquor. The starting solution thus gets separated, at this stage into three fractions, as shown in the third row of Fig. 9.1.

The three fractions are then submitted to crystallisation as before and the resulting liquors are combined with crystals from the adjacent solution, as indicated by arrow heads. The four fractions which thus result, on being processed in the same way as before, give rise to five fractions. Each series of operations, therefore, gives one more fraction.

As a result of the repetition of the above process, the components which are less soluble continuously pass on to the crop of crystals, while those which are soluble pass to the final mother liquor. The process is continued, until the desired degree of separation has been achieved. The systematic fractional crystallisation of any given mixture of rare-earths results in the separation into:

(a) the less soluble components that accumulate more and more in each successive crop of crystals as shown on the left-hand side of Fig. 9.1.

(b) the more soluble components that progressively accumulate in the mother liquor as shown on the right-hand side of Fig. 9.1.

(c) the intermediate components, distributed between the final crop of crystals and the end mother liquor, in the order of solubility, i.e., from left to right as shown at the bottom row of Fig. 9.1.

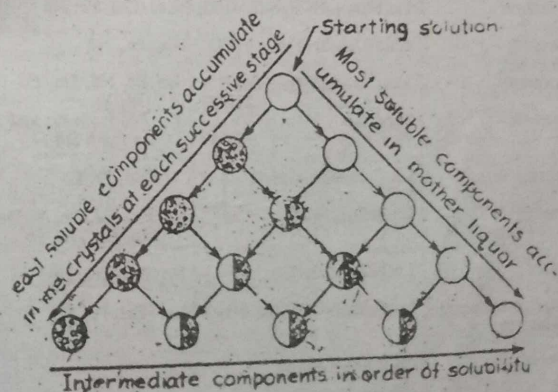


Fig. 9.1. Fractional crystallisation method for the separation of rare-earths.

(ii) **Fractional Precipitation Method.** We have already pointed out that the basic character of the hydroxides of lanthanide elements decreases from La_{57} to Lu_{71} . Thus, the hydroxides of cerium group lanthanide elements (La_{57} – Eu_{63}) are more basic than those of ytterbium group lanthanides (Gd_{64} – Lu_{71}). When pH of the solution containing mixture of the lanthanide elements is gradually increased by adding NH_3 , amines or alkali etc., the hydroxides of the ytterbium group lanthanides being less basic get preferentially precipitated while those of the cerium group lanthanides which are more basic are left behind in the solution. From a given sub-group individual lanthanides are then separated by the further fractional precipitation of the hydroxides. Since the basicity differences in the hydroxides of lanthanide elements are small, complete separation of the adjacent members in a given sub-group by this method of fractional precipitation cannot be made.

(iii) **Fractional Thermal Decomposition of Oxy-salts.** This method is based on the fact that the rate of thermal decomposition of oxy-salts such as nitrates, sulphates and acetates of lanthanide elements at a given temperature decrease from La_{57} to Lu_{71} . This in other words means that the oxy-salts of cerium group elements decompose more readily than those of the ytterbium group. Thus if a mixture containing the nitrates of lanthanide elements is fused and then leached with water, the ytterbium group lanthanides which are less basic are rapidly concentrated and their separation from the more basic cerium lanthanides is effected. The individual members of a given sub-group can be separated from each other by repeating the whole process several times.

Table 9-2. Chief salts used in the fractional crystallisation procedures.

Salts	Composition	The order of separation
(1) Double Nitrates		
Ammonium	$\text{Ln}(\text{NO}_3)_3 \cdot 2\text{NH}_4\text{NO}_3 \cdot 4\text{H}_2\text{O}$	La, (Ce), Pr, Nd
Sodium	$\text{Ln}(\text{NO}_3)_3 \cdot 2\text{NaNO}_3 \cdot x\text{H}_2\text{O}$	La, Pr, Nd
Magnesium	$2\text{Ln}(\text{NO}_3)_3 \cdot 3\text{Mg}(\text{NO}_3)_2 \cdot 24\text{H}_2\text{O}$	La, Pr, Nd, Sm, Eu, Er, Gd, Y
Manganese	$2\text{Ln}(\text{NO}_3)_3 \cdot 3\text{Mn}(\text{NO}_3)_2 \cdot 24\text{H}_2\text{O}$	For Nd from Pr, and for La from Sm
(2) Ethyl sulphates	$\text{Ln}(\text{C}_2\text{H}_5\text{SO}_4)_3 \cdot 9\text{H}_2\text{O}$	(i) Nd, Pr, La
(3) Bromates	$\text{Ln}(\text{BrO}_3)_3 \cdot 9\text{H}_2\text{O}$	(ii) Eu, Gd, Tb, Y, Dy, Ho, Er, Tm, Yb
(4) Nitrates	$\text{Ln}(\text{NO}_3)_3 \cdot 5\text{H}_2\text{O}$	Nd, Pr, (Ce), La
(5) Metanitro-benzene Sulphonates	$\text{Ln}[\text{C}_6\text{H}_4(\text{NO}_2)\text{SO}_3]_3 \cdot 6\text{H}_2\text{O}$	La, (Ce), Pr, Nd
(6) Oxalates	$\text{Ln}_2(\text{C}_2\text{O}_4)_3 \cdot x\text{H}_2\text{O}$	La
(7) Sulphates	$\text{Ln}_2(\text{SO}_4)_3 \cdot x\text{H}_2\text{O}$	La, Nd, Pr
(8) Double carbonates	$\text{K}_2\text{CO}_3 \cdot \text{Ln}_2(\text{CO}_3)_3 \cdot 12\text{H}_2\text{O}$	La, Pr, (Ce), Nd

Since the differences in the basicity of the lanthanides are small, complete separation of the adjacent members of a given sub-group cannot be achieved by this method.

(iv) **Change of oxidation states by selective oxidation or reduction procedures.** As already discussed, some of the lanthanides show +2 and +4 oxidation states in addition to +3 state which is the most characteristic of all the elements of lanthanide series. The properties of Ln^{2+} and Ln^{4+} ions are different from those of the usual Ln^{3+} ions. A change in oxidation state results in a remarkable change in properties. Selective oxidation or reduction, therefore, provides a very effective method of separating specific lanthanides.

Examples : (i) Suppose we have to separate Ce from a mixture containing Ce^{3+} and other tripositive lanthanide cations. This mixture is treated with some strong oxidising agent such as bromate in neutral medium or chlorine or KMnO_4 in alkaline medium. By this treatment only Ce^{3+} ion is oxidised to Ce^{4+} ion while other lanthanide ions remain as Ln^{3+} ions. Since Ce^{4+} ion is smaller than Ce^{3+} ion, Ce^{4+} ion is less basic and less soluble. It (i.e. Ce^{4+} ion) can thus be precipitated as $\text{Ce}(\text{OH})_4$ by the addition of a small amount of an alkali leaving all other Ln^{3+} ions in solution. By this method it has been possible to have 99% pure Ce from a mixture containing only 40% Ce.

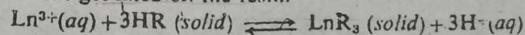
(ii) Eu^{3+} can almost quantitatively be separated from a solution containing Eu^{3+} and other Ln^{3+} ions by reducing it to Eu^{2+} by means of Zn-amalgam and then precipitating it as EuSO_4 , which is insoluble in water and hence can be separated. The sulphates of all the trivalent lanthanides, on the other hand, are soluble in water.

Zn-amalgam also brings about the reduction of Yb^{3+} to Yb^{2+} but not of Sm^{3+} .

(2) Modern Methods.

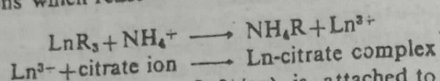
Following are the modern methods which are now used for separation of individual lanthanides.

(i) **Ion-exchange method.** This is the most effective method now available for the separation of lanthanide elements. This method utilises synthetic cation-exchange resins which are usually organic compounds containing sulphonic ($-\text{SO}_3\text{H}$) or carboxylic acid ($-\text{COOH}$) groups, the hydrogens of which are replaced by other cations. Thus, if an aqueous solution containing a mixture of tripositive lanthanide ions, $\text{Ln}^{3+}(\text{aq})$ is allowed to pass down a column packed with a cation exchange resin [abbreviated as $\text{HR}(\text{solid})$], the $\text{Ln}^{3+}(\text{aq})$ ions replace H^+ ions of $-\text{SO}_3\text{H}$ or $-\text{COOH}$ group of the resin and thus get fixed on the resin.



Thus all the $\text{Ln}^{3+}(\text{aq})$ ions are attached with the resin as $\text{LnR}_3(\text{solid})$. Since $\text{Lu}^{3+}(\text{aq})$ is largest in size and $\text{La}^{3+}(\text{aq})$ is smallest, $\text{La}^{3+}(\text{aq})$ is attached to the resin in the column with maximum and $\text{Lu}^{3+}(\text{aq})$ with minimum firmness. In order to remove Ln^{3+} ions fixed at $\text{LnR}_3(\text{solid})$ on the resin, the column is eluted (i.e. leached) with a complexing agent in aqueous solution like a buffer solution of

ammonium citrate-citric acid ($\text{pH} = 4$ to 7). Such complexing agents are called eluants or eluates or eluting agents. During elution process NH_4^+ ions of the eluting agent replace Ln^{3+} ions from LnR_3 (solid) to give Ln^{3+} ions which react with citrate ion to form the Ln-citrate complex.



We have seen that since $\text{La}^{3+}(\text{aq})$ is attached to resin with maximum and $\text{Lu}^{3+}(\text{aq})$ with minimum firmness, Lu-citrate complex comes out of the column first and La-citrate complex comes out last.

In actual practice the process of elution is to be repeated several times by careful control of concentration of ammonium citrate-citric acid solutions.

The yttrium group rare-earths have been separated by Kettle and Boyd on a column of Dowex 50 (270–325 mesh) using 5% (0.25 M) citric acid at $\text{pH} 3.28$. Mixtures of Eu–Sm and Y–Tb took eight hours for complete separation with 0.5 M citrate at $\text{pH} 3.04$ (87°C) and with 0.24 M lactate at $\text{pH} 5.0$. Cunningham *et al.* reduced the separation time to four hours using 1 M lactate ($\text{pH} = 3.25$) at 87°C for a mixture of Y–Pr–Nd–Pm–Sm–Eu with a Zeokarb 225.

Elution Curves. The elution behaviour of each of the constituent rare-earth ions is shown as an elution plot which is obtained by plotting volume of eluted portion against the concentration of the cation in eluate. The elution curves for the elements, Lu to Eu, present in a mixture are shown in Fig. 9.2. These elution curves follow one another, from left to right, in the order of decreasing atomic number.

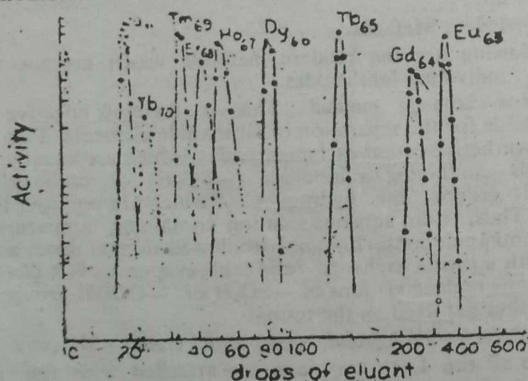
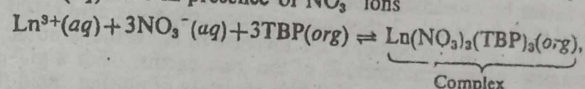


Fig. 9.2. A typical elution plot for the separation of rare-earths (Lu to Eu)

(ii) Solvent (liquid-liquid) extraction method. The separation of lanthanides by this method was first reported by Fischer. The method is based on the differences in the solubility of lanthanide

salts in water and an immiscible or partially miscible organic solvent. These organic solvents are called extracting solvents or extractants. This method has been used on both tracer and macroscales. The technique consists of passing the aqueous solution of lanthanide salts counter-current to the organic solution which extracts the lanthanides from water. The most widely used extracting solvent is tri-n-butyl phosphate, TBP in an inert medium like kerosene or xylene to extract the lanthanides from nitric acid solutions. TBP forms complexes with $\text{Ln}^{3+}(\text{aq})$ ions in presence of NO_3^- ions



where (org) represents the organic phase. The distribution between these two phases (i.e. solvents) is described by distribution ratio (also called extraction coefficient), λ , given by

$$\lambda = \frac{\text{Total concentration of solute in one solvent}}{\text{Total concentration of solute in the other solvent}}$$

$$= \frac{C_{\text{Ln}(\text{NO}_3)_3(\text{TBP})_3(\text{org})}}{C_{\text{Ln}^{3+}(\text{aq})}} \quad \dots(9.1)$$

For two tripositive lanthanide ions, Ln^{3+} and Ln^{2+} , the separation factor, α , is given as,

$$\alpha = \frac{\lambda'}{\lambda} = \frac{C_{\text{Ln}(\text{NO}_3)_3(\text{TBP})_3(\text{org})} \cdot C_{\text{Ln}^{2+}(\text{aq})}}{C_{\text{Ln}(\text{NO}_3)_3(\text{TBP})_3(\text{org})} \cdot C_{\text{Ln}^{3+}(\text{aq})}} \quad \dots(9.2)$$

Peppard has reported that an average separation factor for adjacent lanthanides for 15.8 M nitric acid–100% TBP system is about 1.5.

Equilibrium constant, K , is given by :

$$K = \frac{C_{\text{Ln}(\text{NO}_3)_3(\text{TBP})_3(\text{org})}}{C_{\text{Ln}^{3+}(\text{aq})} \cdot C_{\text{NO}_3^-}^3(\text{aq}) \cdot C_{\text{TBP}(\text{org})}^3} \quad \dots(9.3)$$

Combination of equation (9.3) with equation (9.1) gives equation (9.4), viz.

$$K = \frac{\lambda}{C_{\text{NO}_3^-}^3(\text{aq}) \cdot C_{\text{TBP}(\text{org})}^3}$$

or $\lambda = K \cdot C_{\text{NO}_3^-}^3(\text{aq}) \cdot C_{\text{TBP}(\text{org})}^3 \quad \dots(9.4)$

Kilogram quantities of 95% pure lanthanides have been prepared by solvent extraction technique. Another organic solvent which is a better extractant than TBP is Di (2-ethyl hexyl) phosphoric acid. For this extractant the average separation factor of 2.5 for adjacent lanthanides has been reported. Other extractants used in the separation process are : monoalkylphosphoric acids : $\text{RP}(\text{O})(\text{OH})_2$ or $\text{R}_2\text{H}_2\text{PO}_4$, pyrophosphate esters : $(\text{RO})_2\text{P}(\text{O})-\text{O}-\text{P}(\text{O})(\text{OR})_2$, phosphonates : $(\text{RO})_2(\text{R})\text{PO}$ and $(\text{RO})(\text{R})\text{P}(\text{O})(\text{OH})$ and 1, 3-diketones or 8-quinolins.

Present status of solvent extraction technique. Although solvent extraction technique is promising as a general technique, it cannot compete with ion exchange technique. The major uses of the solvent extraction process are that it can be used in separating the Ln^{3+} ions from Ln^{4+} ions such as Ce^{4+} and Th^{4+} and in the purification of Ce, Th and La. La is somewhat less readily extracted than other Ln^{3+} ions and it can thus be separated from them rather readily.

(iii) **Paper chromatography.** Cerrai and Testa separated some rare-earths by ascending paper chromatography, using CRL/I papers treated with 0.1M dil. (2-ethylhexyl) orthophosphoric acid (HDEHP). Under the same conditions an increase in R_F value is noted with decreasing atomic number, Z, of the rare-earths. For a given element the R_F value increases with the molarity of eluting acid (HCl) and with temperature, but decreases with the concentration of HDEHP. The following R_F values for 1 M HCl eluant have been reported:

$\text{Ln}^{3+} \rightarrow$	La	Ce	Pr	Sm	Eu	Gd	Dy	Er	Yb	Y	Sc
$R_F \rightarrow$	0.92	0.91	0.90	0.76	0.64	0.51	0.12	0.06	0.03	0.07	0.0

Later, Cerrai and Triulzi used a mixture of 3M HNO_3 and 80% CH_3OH as solvent on Whatman cellulose paper DE-20 to separate La, Ce, Pr, Nd, Sm and Gd. Whatman DE-20 is a cellulose anion exchanger, containing amine groups bonded to the cellulose matrix.

It has been found that a mixture of HCl and various aliphatic alcohols can successfully be used as eluant for the separation of rare-earths by paper chromatography (Whatman No. 1).

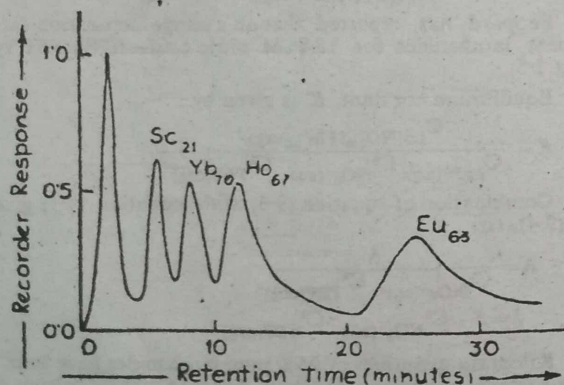


Fig. 9-3. Gas chromatographic separation of a mixture of $\text{Ln}(\text{THD})_3$ complexes in benzene solution [Reproduced from Journal of the American Chemical Society, 87, 5254 (1965)].

(iv) **Gas Chromatography.** The 2, 2, 6, 6-tetra ethyl-3, 5-heptanedione complexes of Ln^{3+} ions, $\text{Ln}(\text{THD})_3$ are quite stable,

volatile and monomeric in C_6H_6 . These complexes have appreciable and significant differences in the volatilities.

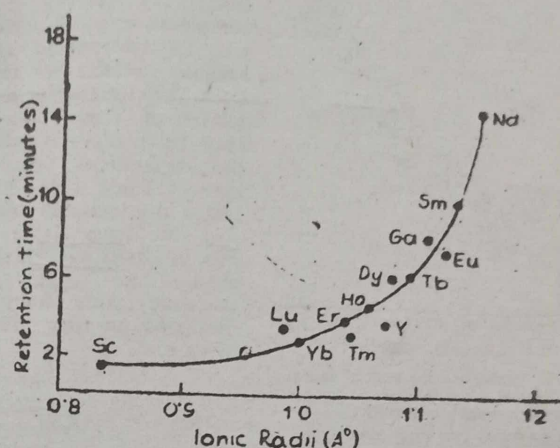


Fig. 9-4. Variation of the retention times (minutes) of the rare-earth tris-complexes with ionic radii ($\text{\AA}$) showing the effect of lanthanide contraction.

Eisenbraut and Sievers have recently separated these complexes by gas chromatographic technique. Fig. 9-3 shows a gas chromatographic separation of a mixture of $\text{Ln}(\text{THD})_3$ complexes in C_6H_6 solution. It has been found that these complexes are eluted without decomposition.

The variation of the retention times (in minutes) of the rare-earth tris-complexes with ionic radii (in $\text{\AA}$) shown in Fig. 9-4 exhibits the effect of lanthanide contraction.

(v) **Thin layer chromatography.** Dancels has separated Sm, Eu, Gd and Tb by this technique using Silica gel H. Radioactive traces were used to detect the rare-earths. The R_F values for Sm, Eu, Gd and Tb are 0.54, 0.29, 0.73 and 1.00 respectively.

(vi) **Complex formation: oxalate precipitation in presence of EDTA.** In the presence of complexing reagent namely ethylenediamine tetra-acetic acid (EDTA), J.K. Marsh (1950-51) has been able to effect an efficient separation of rare-earths by precipitation as oxalates.

Lanthanide-oxalates are insoluble in water, but they can be dissolved by complexing with EDTA.

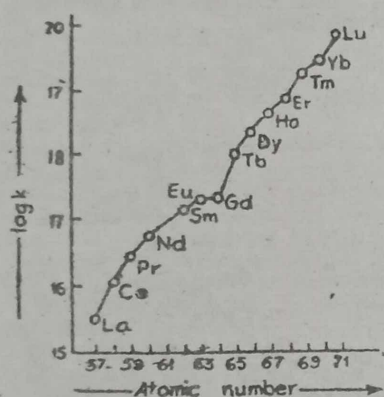


Fig. 9-5. Stability constants of lanthanide-EDTA complexes in aqueous solutions at 20°C.

PREPARATION OF THE METALS

Because of the ease of oxidation of these metals it is not a simple task to obtain the elements in the metallic state.

The following methods may be used for the preparation of metals:

(i) **Thermal decomposition of lanthanide-mercury amalgam.** Using a mercury cathode an amalgam of the lanthanides may be prepared by the electrolysis of aqueous solutions. The amalgam thus formed is thermally decomposed to get the metals in a finely divided state, but complete elimination of mercury is difficult.

(ii) **Electrolysis of fused chlorides.** The anhydrous lanthanide (+3) chlorides are fused with sodium, potassium or calcium chloride by external heating and the molten mass is electrolysed in an iron pot or in a refractory lined pot which serves as the anode. Graphite cathodes are used. The molten metal collects as a pool at the bottom of the pot from where it is withdrawn and cast into ingots.

This method does not give the metals in high degree of purity required in modern technology.

(iii) **Metallurgical reduction of the salts.** In this method the anhydrous fluorides and chlorides of lanthanides are reduced with calcium in a tantalum container at about 1000°C. Calcium is volatilised from the lanthanide-calcium alloy. This method is successful for all the lanthanides except Sm, Eu and Yb, because these metals form stable divalent fluorides. However these metals can be prepared by reducing their oxides with lanthanum metal at 1400°C in vacuum and distilling out Sm, Eu and Yb. These metals are more volatile than lanthanum.

The metals can be purified by vacuum distillation (pressure $\approx 10^{-5}$ mm Hg) in tantalum apparatus.

These lanthanide-EDTA complexes differ in their stabilities. The stability of these complexes, as a rule, increases with an increase in atomic number as shown in Fig. 9-5. The difference in stabilities of these complexes may be made the basis of the separation of rare-earths. Thus, if these complexes are treated with an acid, the least stable complex is destroyed first and changes to oxalate, while the most stable complex is destroyed on the addition of an acid.

GENERAL PROPERTIES OF LANTHANIDES

Some of the important properties of lanthanides are discussed below.

(1) Electronic Configuration

We know that at Ba_{56} 6s-orbital is completely filled ($Ba_{56} \rightarrow [Xe] 6s^2$) and at La_{57} 5d-orbital is singly occupied ($La_{57} \rightarrow [Xe] 5d^1 6s^2$). But after La_{57} further filling of 5d-orbitals is discontinued. As the nuclear charge increases by one unit from La_{57} to Ce_{58} , 4f-orbitals, which were higher in energy up to La_{57} , fall slightly below the 5d level. 4f-orbitals, therefore, begin to fill and are completely filled up to Lu_{71} before the filling of 5d-orbitals is resumed. Electronic configurations of lanthanides are shown in Table 9-3. From this table the following points may be noted:

Table 9-3. Electronic configuration of the neutral gaseous atoms of lanthanides.

Elements with atomic numbers	Complete configurations	Valence-shell configurations
La_{57}	$[Xe] 4f^0 5d^1 6s^2$	$4f^0 5d^1 6s^2$
Ce_{58}	$[Xe] 4f^2 5d^0 6s^2$	$4f^2 5d^0 6s^2$
Pr_{59}	$[Xe] 4f^3 5d^0 6s^2$	$4f^3 5d^0 6s^2$
Nd_{60}	$[Xe] 4f^4 5d^0 6s^2$	$4f^4 5d^0 6s^2$
Pm_{61}	$[Xe] 4f^5 5d^0 6s^2$	$4f^5 5d^0 6s^2$
Sm_{62}	$[Xe] 4f^6 5d^0 6s^2$	$4f^6 5d^0 6s^2$
Eu_{63}	$[Xe] 4f^7 5d^0 6s^2$	$4f^7 5d^0 6s^2$
Gd_{64}	$[Xe] 4f^7 5d^1 6s^2$	$4f^7 5d^1 6s^2$
Tb_{65}	$[Xe] 4f^9 5d^0 6s^2$	$4f^9 5d^0 6s^2$
Dy_{66}	$[Xe] 4f^{10} 5d^0 6s^2$	$4f^{10} 5d^0 6s^2$
Ho_{67}	$[Xe] 4f^{11} 5d^0 6s^2$	$4f^{11} 5d^0 6s^2$
Er_{68}	$[Xe] 4f^{12} 5d^0 6s^2$	$4f^{12} 5d^0 6s^2$
Tm_{69}	$[Xe] 4f^{13} 5d^0 6s^2$	$4f^{13} 5d^0 6s^2$
	$[Xe] 4f^{14} 5d^0 6s^2$	$4f^{14} 5d^0 6s^2$
Lu_{71}	$[Xe] 4f^{14} 5d^1 6s^2$	$4f^{14} 5d^1 6s^2$

18, 8 $[Xe]$ indicates the complete configuration of Xe atom which is 2, 8, 18,

(i) The complete electronic configuration of lanthanides can be represented by a general configuration viz., 2, 8, 18, $4f^0 4f^{14} 5d^0 5d^1 6s^2$.

$5s^2 4d^0$ or $1, 6s^2$ (including La) which shows that the valence-shell configuration can be represented as $4f^0, \dots, 5d^0$ or $1, 6s^2$.

The valence-shell configuration indicates that the additional electron enters the 4f-orbital without altering the electrons in the outermost 6s-orbital.

(ii) The filling of 4f-orbitals is not regular, e.g., the additional electron in Gd_{44} does not enter 4f-level but, instead, it goes to 5d-level. This is because the 4f and 5d orbitals in Gd are at about the same energy level and Gd atom has a tendency to retain the configuration with half-filled 4f-orbitals which are relatively more stable.

(2) Oxidation states and oxidation potentials.

Observed oxidation states of lanthanides noted either in solution or in insoluble compounds are given in Table 9-4 from which it may be noted that whatever the electronic configurations of the lanthanides in the ground state, all of the lanthanides form the tripositive lanthanide cations. This fact is not directly evident from the electronic configurations, it is actually due to the fact that the magnitudes of the energy required to remove an electron from the gaseous ion in its lower oxidation state (i.e., ionisation energy) and of that released when two gaseous ions combine with water to form the aquated species (i.e., hydration energy) are such that all the tetrapositive species (except possibly Ce^{4+}) and all the dipositive species (except Eu^{2+}) revert to the tripositive species. Thus this leads to the conclusion that tripositive species are more stable than the di- and tetrapositive species in aqueous solutions.

In the solid state too the combination of ionization energy and the energy released when gaseous ions combine to produce crystalline solids (i.e., the lattice or crystal energy) is more negative for the tripositive species than for di- and tetrapositive species. Consequently, the tripositive lanthanides are also the most common in the solid compounds.

Stability of various oxidation states. It is possible to correlate the stability of various oxidation states of lanthanides with the electronic configuration of their ions. On the basis of general rule that empty, half-filled and completely filled 4f orbitals are highly stable, the formation of Ce^{4+} , La^{3+} ($4f^0$), Tb^{4+} , Eu^{3+} , Gd^{3+} ($4f^7$) and Yb^{2+} , Lu^{3+} ($4f^{14}$) ions can be explained. It is, however, difficult to explain the stability of oxidation states of the cations other than those given above. It may thus be assumed that in addition to the special stability associated with $4f^0$, $4f^7$ and $4f^{14}$ configurations, there may be other factors such as thermodynamic and kinetic, in determining the stability of various oxidation states of lanthanides.

The stability order of +2 state is $Eu > Yb > Sm > Tm \sim Nd$. Ln solids ($Ln = La, Ce, Pr$ and Gd) do not contain Ln^{2+} ions, but are metallic in nature. The stability order of +4 state is $Ce > Tb \sim Pr > (Nd \sim Dy)$.

The ease of formation of the various oxidation states in solution is indicated by the values of the standard electrode potential, E° .

These values (in volts) for different couples of lanthanides such as $Ln^0 \rightarrow Ln^{3+} + 3e^-$, $Ln^{2+} \rightarrow Ln^{3+} + e^-$ and $Ln^{3+} \rightarrow Ln^{4+} + e^-$, for 1M perchloric acid at 25°C are tabulated in Table 9-4. Here Ln represents the elemental lanthanide. Ln^{2+} , Ln^{3+} and Ln^{4+} are the di-, tri- and tetrapositive lanthanide ions. From this table, we observe that:

(i) The high positive values of the oxidation electrode potentials for the couple $Ln^0(s) \rightleftharpoons Ln^{3+}(aq) + 3e^-$ indicate that the elemental lanthanides are powerful reducing agents, i.e. oxidation of the lanthanide metals to the tripositive state occurs readily and vigorously. The gradual decrease, though very slow, in the values of E° indicates very slight decrease in chemical activity from one element to the next one.

(ii) The enhanced stabilities associated with the empty, half-filled and completely filled 4f-orbitals is also indicated by these values. Thus Ce^{4+} ($4f^0$) is much less readily reduced to the tripositive ion, Ce^{3+} ($4f^1$) than Pr^{4+} ion ($4f^1$). Furthermore, the 4f⁷ species (e.g. Eu^{3+} ion) and the 4f¹⁴ species (e.g. Yb^{3+} ion) are the weakest reducing agents of the di-positive species.

(iii) The values of E° for couple $Ln^0(s) \rightarrow Ln^{3+}(aq) + 3e^-$ decreases with the increase of atomic number, as is evident from Table 9-4. The high values of E° are in accordance with the electropositive character of lanthanides. The decrease in the values of E° with the increase in atomic number (i.e. a decrease in the electropositive character) is evidently inconsistent with the corresponding decrease in the ionic radius, i.e. lanthanide contraction (to be described subsequently).

Chemistry of +3 state. Long

In general, the lanthanides behave as active metals. Their electrode potential values are comparable to those of alkaline earths. All the metals act as strong reducing agents. They are all attacked by acids with the liberation of hydrogen. The metals dissolve slowly in cold but more rapidly in warm water and liberate hydrogen. They absorb hydrogen to form interstitial hydrides.

Nearly all known anions form the compounds with Ln^{3+} cation. These compounds are stable in solid as well as in solution state. Compounds of Ln^{3+} cation with the anions such as OH^- , CO_3^{2-} , SO_4^{2-} , $C_2O_4^{2-}$, NO_3^- etc. decompose on heating, give first basic salts and finally oxides. Hydrated salts that contain thermally stable anions such as F^- , Cl^- , Br^- , PO_4^{3-} etc. also give similar products on heating because of hydrolysis.

Compounds of Ln^{3+} cation with the anions Cl^- , Br^- , I^- , NO_3^- , CH_3COO^- , BO_3^{3-} are generally soluble in water while those with F^- , OH^- , O^{2-} , $C_2O_4^{2-}$, CO_3^{2-} , CrO_4^{2-} , PO_4^{3-} etc. are generally insoluble.

Oxides, Ln_2O_3 . The oxides, Ln_2O_3 , are formed by heating the metal in oxygen or by the thermal decomposition of the $Ln(OH)_3$ or oxy salts such as $Ln_2(CO_3)_3$ and $Ln(NO_3)_3$. The oxides resemble those of alkaline earth oxides. All the oxides are almost insoluble

Lanthanides or Rare Earths (4f-Block Elements)

Table 9.4. Various oxidation states shown by lanthanides and standard oxidation potentials, E° (in volts) for various couples of lanthanide elements in 1M perchloric acid at 25°C

Lanthanide elements	Oxidation states (Less stable states are shown in brackets)	E° values (in volts) for various couples (Estimated values are given in brackets)
La ₅₇	+3	La ⁰ /La ³⁺ = 2.52 volts
Ce ₅₈	+3, (+4)	Ce ⁰ /Ce ³⁺ = 2.48
Pr ₅₉	+3, +4	Pr ⁰ /Pr ³⁺ = 2.46, Pr ³⁺ /Pr ⁴⁺ = -1.74
Nd ₆₀	+2, +3	Nd ⁰ /Nd ³⁺ = 2.43, Nd ³⁺ /Nd ⁴⁺ = (-2.86)
Pm ₆₁	+3	Pm ⁰ /Pm ³⁺ = (2.42)
Sm ₆₂	(+2), +3	Sm ⁰ /Sm ³⁺ = 2.41, Sm ²⁺ /Sm ³⁺ = 1.55
Eu ₆₃	(+2), +3	Eu ⁰ /Eu ³⁺ = 2.40, Eu ²⁺ /Eu ³⁺ = 0.43
Gd ₆₄	(+2), +3	Gd ⁰ /Gd ³⁺ = 2.39
Tb ₆₅	+3, (+4)	Tb ⁰ /Tb ³⁺ = 2.39
Dy ₆₆	+3, +4	Dy ⁰ /Dy ³⁺ = 2.35
Ho ₆₇	+3	Ho ⁰ /Ho ³⁺ = 2.32, Ho ²⁺ /Ho ³⁺ = 0.57
Er ₆₈	+3	Er ⁰ /Er ³⁺ = 2.30
Tm ₆₉	(+2), +3	Tm ⁰ /Tm ³⁺ = 2.28
Yb ₇₀	(+2), +3	Yb ⁰ /Yb ³⁺ = 2.27, Yb ²⁺ /Yb ³⁺ = 1.15
Lu ₇₁	+3	Lu ⁰ /Lu ³⁺ = 2.25

in water. They absorb CO₂ and H₂O from air to form carbonates and hydroxides respectively.

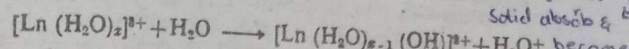
Hydroxides, Ln(OH)₃. The hydroxides are precipitated as gelatinous precipitates from aqueous solution by the addition of ammonia or dil. alkali to soluble salts of Ln³⁺ ion in solution. The precipitation of these hydroxides on the addition of an alkali or ammonia to their salts is as : Sc, Lu, Yb, Tm, Er, Ho, Dy, Tb, Sm, Gd, Eu, Y, Nd, Pr, Ce, La.

These hydroxides are not amphoteric. These are definite compounds with hexagonal structure and are not merely hydrous oxides. They absorb CO₂ to give normal carbonates.

The oxides and hydroxides are basic. The basicity decreases with increasing atomic number. Thus La₂O₃ and La(OH)₃ are the most basic while Lu₂O₃ and Lu(OH)₃ are least basic. La₂O₃ is a strong base and slakes like CaO on addition of water. The hydroxides are decomposed on heating to form oxides.

Lanthanides or Rare Earths (4f-Block Elements) but remain soluble e.g. NaCl

Oxy salts. Lanthanide salts of most oxy acids (called oxy salts) such as sulphates, nitrates, perchlorates, bromates etc. are also known. These are generally soluble and crystallise as hydrates. Soluble oxy salts such as sulphates, nitrates and perchlorates can be prepared by dissolving the oxides, hydroxides or carbonates in the appropriate oxyacids. From the solution, hydrated salts can be crystallised out. The nitrates are often deliquescent and crystallise with 6H₂O. Solutions of oxy salts furnish hydrated cations, [Ln(H₂O)₉]³⁺ which tend to undergo slight hydrolysis in aqueous solution:



The smaller the ionic radius of Ln³⁺ ion, the greater is the tendency of the ion, [Ln(H₂O)₉]³⁺, to hydrolyse. Thus the tendency of the [Ln(H₂O)₉]³⁺ ion to hydrolyse increases with increasing atomic number, since on passing from La³⁺ to Lu³⁺ there is a contraction in their ionic radii.

Double Salts. Lanthanide salts form a large number of double salts. The most important double salts are:

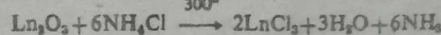
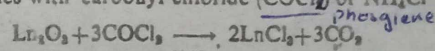
(i) Double nitrates such as 2 Ln(NO₃)₃ · 3M^{II}(NO₃)₂ · 24H₂O (M^{II} = Mg, Zn, Ni, Mn) and Ln(NO₃)₃ · 2NH₄NO₃ · 4H₂O.

(ii) Double sulphates such as Ln₂(SO₄)₃ · 3M^ISO₄ · 12H₂O (M = alkali metal) L₁ → solid.

The double sulphates, Ln₂(SO₄)₃ · 3Na₂SO₄ · 12H₂O where Ln = La—Eu are only sparingly soluble in Na₂SO₄, while those where Ln = Gd—Lu are appreciably soluble. Thus a separation of lanthanides into two groups: Cerium group having La₅₇—Eu₆₃ lanthanides and Yttrium group having Gd₆₄—Lu₇₁ lanthanides is possible. Since the double salts crystallise well, these are used to separate the rare-earths from one another.

Halides, LnX₃. Fluorides are precipitated by the addition of HF or a soluble fluoride to a Ln^{III} salts solution. The fluorides particularly of heavier lanthanides, are sparingly soluble in HF due to the formation of fluoro complexes.

The anhydrous chlorides can be prepared by the direct combination of the elements on heating. These are best prepared by heating the oxides with carbonyl chloride (COCl₂) or NH₄Cl.



The anhydrous chlorides cannot be obtained from the hydrated chlorides, since these lose HCl on heating to give the oxychlorides, (LnOCl), more easily than they lose H₂O. (ScCl₃ · 3H₂O and CeCl₃ · 2H₂O, however, give Sc₂O₃ and Ce₂O₃). The chlorides are non-volatile, deliquescent solids soluble in H₂O and alcohol. The

hydrated chlorides are formed by dissolving the oxides or carbonates in HCl and concentrating the solution to crystallising point. They crystallise from solution usually as hexahydrates, $\text{LnCl}_3 \cdot 6\text{H}_2\text{O}$.

Bromides and iodides are rather similar to the chlorides. Iodides of the first few lanthanides are orthorhombic while those of the remaining lanthanides are hexagonal.

Carbonates, $\text{Ln}_2(\text{CO}_3)_3$: The normal carbonates can be prepared by passing CO_2 into an aqueous solution of $\text{Ln}(\text{OH})_3$. They can also be prepared by adding Na_2CO_3 solution to Ln^{III} salt solution. The carbonates are insoluble in H_2O , but dissolve in acids with liberation of CO_2 and forming Ln^{III} salts.

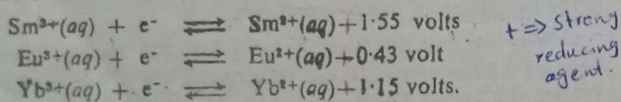
Phosphates and Oxalates: These are also insoluble in water. All lanthanides are quantitatively precipitated as oxalates from Ln^{III} solution containing $\text{C}_2\text{O}_4^{2-}$ ion. The precipitate on drying and ignition gives Ln_2O_3 .

Compounds with non-metals and metalloids: The lanthanides also form well-defined compounds with non-metals and metalloids other than oxygen and halogens. These compounds can generally be prepared by the direct combination of the elements at elevated temperatures. Among these compounds are Ln_2S_3 (not of Eu), selenide (Ln_2Se_3) and oxy sulphides ($\text{Ln}_2\text{O}_3\text{S}$). Nitrides (LnN), phosphides (LnP), arsenides (LnAs), antimonides (LnSb) and bismuthides (LnBi) crystallise in the NaCl structure. The carbide (Ln_2C), is found for the lanthanides, Sm—Lu but not for La, Ce, Pr and Nd.

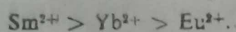
Chemistry of + 2 state.

This is an anomalous oxidation state. The lanthanides showing oxidation state can be divided into +2 two categories:

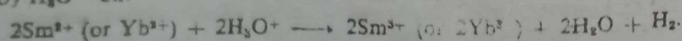
(a) $\text{Sm}_{0.2}$, $\text{Eu}_{0.2}$ and $\text{Yb}_{0.2}$. The dipositive ions of these lanthanides i.e. Sm^{2+} , Eu^{2+} and Yb^{2+} exist in solution. The standard oxidation potentials at 25°C , in acid solution, of these cations are given below:



These values indicate that Sm^{2+} , Eu^{2+} , and Yb^{2+} ions are strong reducing agents and their reducing strength is in the order:



Sm^{2+} and Yb^{2+} ions are rapidly oxidised by H_3O^+ ions (acidic solution), while Eu^{2+} ion is fairly stable and is only slowly oxidised by H_3O^+ ion.

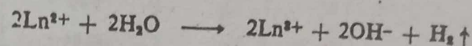


All these cations are rapidly oxidised in presence of oxygen, ^{cos they are itself good reducing agent}
e.g., $4\text{Ln}^{2+} + 4\text{H}_3\text{O}^+ + \text{O}_2 \longrightarrow 4\text{Ln}^{3+} + 6\text{H}_2\text{O}$, where Ln^{2+} may be Sm^{2+} , Eu^{2+} or Yb^{2+} .

The compounds of Sm^{2+} , Eu^{2+} and Yb^{2+} which are insoluble in H_2O are not oxidised by H_2O , while hydrated water-soluble compounds of Sm^{2+} and Yb^{2+} are oxidised by their water. Hydrated water-soluble compounds of Eu^{2+} are more stable.

(b) $\text{Ce}_{0.8}$ (?), $\text{Nd}_{0.8}$ and $\text{Tm}_{0.8}$. The compounds having these elements in +2 oxidation state are known only as solid halides. These are immediately oxidised with air.

(c) Of the divalent compounds of lanthanides, those of Eu^{2+} ion are most stable. The compounds of Ln^{2+} ion are not stable in solution. All the Ln^{2+} compounds decompose water with evolution of H_2 .

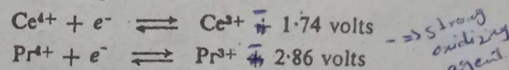


The reaction is, however, sluggish; with Eu the reaction is so retarded that Eu^{2+} compounds may be regarded as fairly stable in aqueous solutions at an ordinary temperature.

Chemistry of +4 state.

This oxidation state is also an anomalous oxidation state. Double salts like $\text{Ce}(\text{NO}_3)_4 \cdot 2\text{NH}_4\text{NO}_3$ and $\text{Ce}(\text{SO}_4)_2 \cdot 2(\text{NH}_4)_2\text{SO}_4 \cdot 2\text{H}_2\text{O}$ have also been prepared.

The standard oxidation potentials at 25°C , in acid solution, of Ce^{4+} and Pr^{4+} ions are given as under:



These values show that $\text{Ce}(\text{IV})$ and $\text{Pr}(\text{IV})$ are strong oxidising agents, the latter being by far the stronger of the two. $\text{Ce}(\text{SO}_4)_2$ is generally used in volumetric analysis. Ce^{4+} ion is readily reduced to Ce^{3+} ion.

The tetravalent ions of Ce are stable in the solid state as well as in solution; Pr^{IV} , Nd^{IV} , Tb^{IV} and Dy^{IV} are stable only in solution.

(3) Atomic and Ionic Radii—Lanthanide Contraction

The atomic radii of the Ln atoms and the crystal or ionic radii of Ln^{2+} , Ln^{3+} , and Ln^{4+} ions are given in Table 9.5.

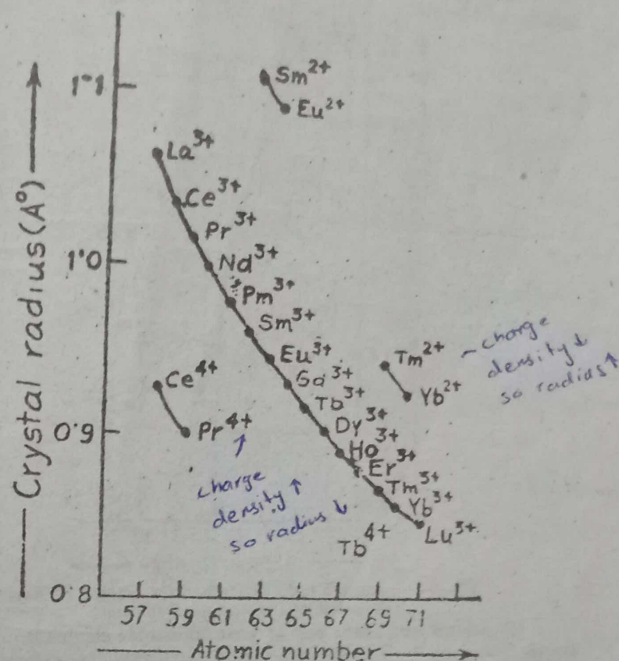
This table clearly shows that as we move along the lanthanide series, there is a decrease in atomic and ionic radii. This steady decrease in the atomic and ionic radii is called lanthanide contraction. This decrease in size is regular in ions (Fig. 9-6) but not so regular in case of atoms (Fig. 9-7).

Table 9-5. Atomic and Ionic (or crystal) Radii of lanthanides and allied elements in Å

Elements with atomic No.	Atomic Radius (r_{Ln})	Crystal or ionic radii of		
		Ln^{2+} ion	Ln^{3+} ion	Ln^{4+} ion
Sc ₂₁	1.64		0.68	
Y ₃₉	1.80		0.88	
La ₅₇	1.88		1.06	
Ce ₅₈	1.88		1.03	0.92
Pr ₅₉	1.83		1.01	0.90
Nd ₆₀	1.82		0.99	
Pm ₆₁	—		(0.98)	
Sm ₆₂	1.80	1.11	0.96	
Eu ₆₃	2.04	1.09	0.95	
Gd ₆₄	1.80		0.94	
Tb ₆₅	1.78		0.92	0.84
Dy ₆₆	1.77		0.91	
Ho ₆₇	1.77		0.89	
Er ₆₈	1.76		0.88	
Tm ₆₉	1.75	0.94	0.87	
Yb ₇₀	1.94	0.93	0.86	
Lu ₇₁	1.73		0.85	

Fig. 9-7 clearly shows that the graph has two peaks, one at Eu₆₃ and the other at Yb₇₀ which possess exceptionally high values for their atomic radii. This is regarded due to the difference in metallic bonding. Each of the lanthanides is assumed to contribute three electrons for the formation of metallic bond, excepting Eu and Yb. These two atoms contribute only two electrons towards bond formation, thus leaving behind half-filled and completely filled 4f-orbitals respectively. The three vertical lines shown in the figure have been drawn just to mark off the lanthanide part of the curve and to divide the lanthanides into two portions marked as 1 and 2. The dividing line passes through Gd₆₄ which has half-filled 4f-shell (Gd $\rightarrow 4f^7 5d^1 7s^2$). The right-hand half of the curve is obviously a duplicate pattern of left-hand half. The seven elements of portion 1 namely from Ce₅₈ to Gd₆₄ occupy a position corresponding to

that occupied by the seven elements namely Tb₆₅ to Lu₇₁ of portion 2.

Fig. 9-6. Crystal radii of Ln^{2+} , Ln^{3+} and Ln^{4+} ions as a function of their atomic numbers.

Cause of lanthanide contraction, poor shielding effect of f-e

We have already seen that in lanthanides the additional electron enters 4f-sub-shell but not in the valence-shell namely sixth shell. The shielding effect of one electron in 4f-sub-shell by another in the same sub-shell (i.e. mutual shielding effect of 4f-electrons) is very little, being even smaller than that of d-electrons, because the shape of f-sub-shell is very much diffused. The nuclear charge (i.e. atomic number), however, increases by unity at each step. Thus the nuclear charge increases at each step, while there is no comparable increase in the mutual shielding effect of 4f-electrons. This results in that electrons in the outermost shell experience increasing nuclear attraction from the growing nucleus. Consequently the atomic and ionic radii go on decreasing as we move from La₅₇ to Lu₇₁.

Consequences of Lanthanide Contraction.
Some important consequences of the lanthanide contraction are summarised below:

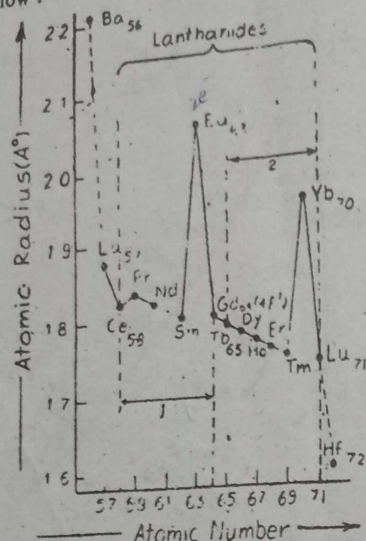


Fig. 9-7 Atomic radii of lanthanides, Ba and Hf as a function of their atomic numbers.

(i) **Anomalous behaviour of post-lanthanide elements.** The elements following lanthanides are called *post-lanthanide elements*. Thus these are the elements of third transition series. Lanthanide contraction effects the properties of post-lanthanide elements. The following examples illustrate this point.

(a) **Atomic and ionic radii of post-lanthanide elements.** Normally in the same sub-group, the covalent radii increase as the atomic number increases. This fact is evident when we compare the covalent radii values of the elements of the first and second transition series as given in Table 9-6.

If we compare the covalent radii values of the elements of the second transition series with those of the third series, we find that the normal increase in the covalent radii values from Sc_{21} to Y_{39} to La_{57} disappears after lanthanides and the pairs of elements: $\text{Zr}-\text{Hf}$, $\text{Nb}-\text{Ta}$, ..., $\text{Ag}-\text{Au}$, $\text{Cd}-\text{Hg}$ possess almost same values. If the fourteen lanthanides had not been intervened between La_{57} and Hf_{72} , the value of covalent radius, for example, of Hf_{72} should have been greater than that of Zr_{40} , since Zr and Hf both are in the same sub-group IV B and Zr lies immediately above Hf . The occurrence of lanthanide contraction cancels the expected increase in the covalent radius value from Zr to Hf . This situation is also with other pairs of elements in the second and third series of transitional elements.

Table 9-6. Atomic (covalent) Radii (in Å) of the elements preceding and following lanthanides.

Transition series	III B	IV B	V B	VI B	VII B	VIII	I B	II B
First	Sc_{21} 1.44	Ti 1.32	V 1.22	Cr 1.18	Mn 1.17	Fe 1.17 Co 1.16 Ni 1.15	Cu 1.17	Zn_{30} 1.25
Second	Y_{39} 1.62	Zr_{40} 1.45	Nb 1.34	Mo 1.30	Tc 1.27	Ru 1.25 Rh 1.25 Pd 1.28	Ag 1.34	Cd_{48} 1.48
Third	La_{57} 1.69	Hf_{72} 1.44	Ta 1.3	W 1.30	Re 1.28	Os 1.26 Ir 1.27 Pt 1.30	Au 1.34	Hg_{80} 1.49

$\text{Ti} = 215 \text{ pm}$

$\text{Zr} = 160 \text{ nm}$

$= 160000 \text{ pm}$

$\text{Hf} = 225 \text{ pm}$

Size always

↑ from top to

bottom in grp

but it decrease

when move Zr

to Hf as due

to lanthanide

contraction

e.g.

$\text{Cu} 128 \text{ pm}$

$\text{Ag} 172 \text{ pm}$

$\text{Au} 166 \text{ pm}$

14 lanthanides

Hg → most dense liquid on earth Density of gold = 19.3 g

Ag → electric conductance (highest)

Lanthanides or Rare Earths (4f-Block Elements)

300 coinage (Cu, Ag, Au) → from top to bottom but

high electrical conductance. Since the pairs of elements of second and third transition series, occurring after lanthanides, (e.g. Zr—Hf, Nb—Ta etc. pairs) have almost similar size (which is due to lanthanide contraction), they are much more close to one another in properties than do the pairs of elements of first and second transition series, e.g., solubilities of their salts are very much similar to one another.

(b) High density of post-lanthanide elements. Because of lanthanide contraction the atomic sizes of the post-lanthanide elements become very small. Consequently the packing of atoms in their metallic crystals become so much compact that their densities are very high. Thus, while the densities of the elements of second transition series are only slightly higher than those of the elements of first series, the densities of the elements of third transition series are almost double to those of the elements of second transition series.

Basic character (ii) Basic character of oxides, Ln_2O_3 and hydroxides, $\text{Ln}(\text{OH})_3$. The chemical properties of an element or an ion depend on the size of the atom or ion. The greater the atomic or ionic radius, greater is the ease with which the species (i.e. atom or ion) will lose electrons. Thus the ease with which a species loses electrons is, the measure of the basicity of the species. Basicity decreases as the ionic radius decreases. Thus, basicity of Ln^{3+} ions may be expected to decrease in the order: $\text{La}^{3+} > \text{Ce}^{3+} > \text{Pr}^{3+} > \text{Nd}^{3+} > \text{Pm}^{3+} > \text{Sm}^{3+} > \text{Eu}^{3+} > \text{Gd}^{3+} > \text{Tb}^{3+} > \text{Dy}^{3+} > \text{Ho}^{3+} > \text{Er}^{3+} > \text{Tm}^{3+} > \text{Yb}^{3+} > \text{Lu}^{3+}$.

These differences in basicity are reflected in (a) thermal decomposition of oxy-salts, e.g. more basic oxy-salts decompose less easily, (i.e. with difficulty) (b) hydrolysis of ions—more basic ions hydrolyse less readily (c) solubilities of salts (d) formation of complexes, and (e) decreasing ease of oxidation of the metals with increasing atomic number—oxidation potential for the couple $\text{Ln} \rightarrow \text{Ln}^{3+} + 3e^-$ regularly goes on decreasing.

Due to lanthanide contraction the decrease in size of Ln^{3+} ions from La^{3+} to Lu^{3+} increases the covalent character (i.e. decreases the ionic character) between Ln^{3+} and OH^- ions in $\text{Ln}(\text{III})$ hydroxides (Fajans rules). Thus $\text{La}(\text{OH})_3$ is the most basic while $\text{Lu}(\text{OH})_3$ is the best basic.

Similarly there is a decrease in the basic strength of the oxides, Ln_2O_3 with the increase of atomic number of Ln atom.

(iii) Small variation in the properties on account of lanthanide contraction allows the separation of lanthanides by the methods based on fractional crystallisation and basicity differences.

(iv) Occurrence of Y with heavy lanthanides. The crystal radii of Y^{3+} and Er^{3+} are equal ($\text{Y}^{3+} = 0.93 \text{ \AA}$ and $\text{Er}^{3+} = 0.96 \text{ \AA}$). This similarity in ionic size of these two cations coupled with the equality in ionic charge ($= +3$ in both the ions) accounts for the invariable occurrence of Y with heavier lanthanides. The marked similarities in the crystal structure, chemical properties and solubility between yttrium compounds and the corresponding ones of the heavier lanthanides make it difficult to separate yttrium from the heavier lanthanides. It is because of these similarities that yttrium is

Lanthanides or Rare Earths (4f-Block Elements)

Ag has due to lanthanide contraction.

301

regarded for all practical purposes as a member of lanthanide series.

(4) Colour and Absorption Spectra of Ln^{3+} Ions

The striking colour characteristics of crystalline salts of a number of Ln^{3+} ions persist in aqueous and non-aqueous solutions and remain unaffected by the change of the anion present or by the addition of colourless complexing groups. Thus it can be said that the colours are characteristic of Ln^{3+} ions themselves. Colours of Ln^{3+} ions are given in Table 9.7.

From Table 9.7 the following points may be noted :

(i) Ln^{3+} ions having x electrons and $(14-x)$ electrons in $4f$ orbital have the same colour. For example Pr^{3+} and Tm^{3+} ions having 2 and $(14-2)=12$ electrons in the $4f$ -orbital respectively have the same colour namely green. Thus we see that $4f$ -orbitals are the source of colours of Ln^{3+} ions.

(ii) The colours of Ln^{3+} ions from La^{3+} to Gd^{3+} repeat themselves at least qualitatively, from Lu^{3+} back to Gd^{3+} . It can thus be concluded that the colours depend on the number of unpaired electrons in $4f$ -orbitals, but the fact that Ln^{3+} (e.g. Sm^{3+} , Eu^{3+} , Yb^{3+} ions) and Ln^{4+} ions (e.g. Ce^{4+} ion) which are iso-electronic with Ln^{3+} ions have different colours suggests that the situation is somewhat more complex. For example, although both Eu^{3+} ($4f^6$) and Sm^{2+} ($4f^6$) ions have the same number of unpaired electrons in $4f$ -orbitals ($=6$), they have different colours (Eu^{3+} —colourless, Sm^{2+} —reddish). Other examples of pairs of cations having the same number of unpaired electrons in $4f$ -orbitals but with different colours are :

- (a) $\text{La}^{3+}(4f^0, n=0)$ —colourless, $\text{Ce}^{4+}(4f^0, n=0)$ —orange-red
- (b) $\text{Gd}^{3+}(4f^7, n=7)$ —colourless, $\text{Eu}^{2+}(4f^7, n=7)$ —straw-yellow
- (c) $\text{Lu}^{3+}(4f^{14}, n=0)$ —colourless, $\text{Yb}^{2+}(4f^{14}, n=0)$ —Green.

(iii) The colour of the Ln^{3+} cations can, to a large extent, be correlated with the electronic configuration of Ln^{3+} ions. Thus, as is evident from Table 9.7, the Ln^{3+} ions having no unpaired $4f$ -electrons (i.e. having either $4f^0$ or $4f^{14}$ electronic configuration) or half-filled $4f$ -orbital (i.e. $4f^7$ configuration) are colourless, e.g. all the cations viz $\text{La}^{3+}(4f^0)$, $\text{Gd}^{3+}(4f^7)$ and $\text{Lu}^{3+}(4f^{14})$ are colourless.

Origin of Colour

The colours are due to Laporte-forbidden $f-f$ transitions, i.e. transitions between the J states of $4f^n$ configuration. Hence the absorption bands of Ln^{3+} ions (except Ce^{3+} and Yb^{3+} ions) are very weak but sharp when compared to those of d -block elements. Many of these bands are line-like and become even narrower as the temperature is lowered. These narrow bands appear due to $f-f$ transition and are independent of the nature of the anion present. As $4f$ -electrons lie deep inside the atom, the colours of Ln^{3+} ions are not effected by changing the anion.

The colour one sees is the result of absorption of light of certain wavelengths and transmission of light of other wavelengths.

are octahedral. But some 8, 9 and 10 coordinated complexes have also been prepared.

Bonding between lanthanide ions and coordinating ligands depends primarily on the electronegativity of the bonding atom in the ligand. Bond formation follows the order F^- , OH^- , H_2O , NO_3^- , Cl^- etc., for monodentate ligands. Complex formation with bidentate ligands in the presence of water is usually successful only with ligands that form chelate rings through oxygen atoms.

USES OF LANTHANIDES AND THEIR COMPOUNDS

(A) Uses of the elements

(i) Lanthanides are used in *metallothermic reactions* due to their extraordinary reducing property (Ce is a stronger reducing agent than Al). Lanthanido-thermic processes can yield sufficiently pure Nb, Zr, Fe, Co, Ni, Mn, Y, W, U, B and Si. These metals are also used as de-oxidising agents particularly in the manufacture of Cu and its alloys. *mostly exothermic (e.g. Na with H_2O)*

$\Rightarrow$ (ii) Uses of mish-metals. Alloys of lanthanides are known as mish-metals. The major constituents of mish-metals are Ce (45—*but 50%*), La (25%), Nd (5%) and small quantities of other lanthanide *we* metals and Fe and Ca impurities. *breaks ϵ tensile strength*

Mish-metals are used for the production of different brands of steel like heat resistant, stainless and instrumental steels. The addition of 0.75% of mish-metal to steel raises its yield point and its workability in heated state and improves its resistance to oxidation. *specifically across oxygen*

Mg-alloys containing about 30% misch metal and 1% Zr are useful in making parts of jet engine. When alloyed with 30% iron, it is sufficiently pyrophoric to be useful in lighter flints. *heat light*

(B) Uses of the Lanthanide Compounds

The uses of the compounds of lanthanides can broadly be classified as follows :

(a) Non-nuclear applications. The following uses are important.

(i) Ceramic applications. CeO_2 , La_2O_3 , Nd_2O_3 and Pr_2O_3 are widely used for decolorising glass. Approximately 1% CeO_2 is used in the manufacture of protective transparent glass blocks to be used in nuclear technology because these blocks are not affected by prolonged exposure to radiations. Because lanthanide oxides can absorb *un* ultra-violet rays, these are used as additives in glasses for special purposes, e.g. for making (i) sun-glasses (by adding Nd_2O_3) (ii) goggles for glass blowing and welding work ($Nd_2O_3 + Pr_2O_3$) (iii) glasses protecting eyes from neutron radiation ($Gd_2O_3 + Sm_2O_3$) etc.

The addition of more than 1% CeO_2 to a glass gives it a brown colour. Nd_2O_3 and Pr_2O_3 give respectively red and green colours. ($Nd_2O_3 + Pr_2O_3$) gives a blue colour.

(ii) Refractories. CeS (m.pt. = $2000^\circ C$) is used in the manufacture of a special type of crucibles which are used for melting metals

in a reducing atmosphere at temperatures up to 1800°C . Borides, carbides and nitrides of lanthanides are also used as refractories. *most H₂ used as can bear high temp*

(iii) *abrasives* Abrasives. Lanthanide oxides are used as abrasives for polishing glasses, e.g. the mixture of oxides, CeO_2 (47%), $\text{La}_2\text{O}_3 + \text{Nd}_2\text{O}_3 + \text{Pr}_2\text{O}_3$ (51%) + SiO_2 , CaO , Fe_2O_3 etc. ($\approx 2\%$) which is called *polirite* has been used for polishing glasses. *usage imp*

(iv) *Paints*. Lanthanide compounds are used in the manufacture of lakes, dyes and paints for porcelain, e.g. cerium molybdate gives light yellow colour, cerium tungstate gives greenish blue colour and salts of Nd give red colour. *cerium salts*

(v) *In textiles and leather industries*. Ceric salts are used for dyeing in textile industries and as tanning agents in leather industries. $\text{Ce}(\text{NO}_3)_4$ is used as a mordant for alizarin dyes. Chlorides and acetates of lanthanides make the fabrics water-proof and acid resistant. *category of medicine*

(vi) *In medicine and agriculture*. Dimals which are salicylates of Pr and Nd are used as germicides. Cerium salts are used for the treatment of vomiting and sea-sickness. Salts of Er and Ce increase the red-blood corpuscles and haemoglobin content of blood.

In agriculture lanthanide compounds are used as insecto-fungicides and as trace elements in fertilisers.

(vii) *In lamps*. Salts of La, Ce, Eu and Sm are used as activators of luminophores. They are used in the manufacture of gas mantles, in the coatings of luminescent lamps and for painting the screens of cathode-ray tubes.

(viii) *In analytical chemistry*. $\text{Ce}(\text{SO}_4)_2$ is used as an oxidising agent in volumetric titrations.

Radio-isotopes of lanthanides (e.g. La^{140} , Ce^{143} , Eu^{152} , Tb^{160} etc.) are used in the study of co-precipitation, chromatographic separations etc.

(ix) *Catalytic applications*. Certain compounds of lanthanides are employed for the hydrogenation, dehydrogenation and oxidation of various organic compounds. Cerium phosphate finds use as a catalyst in petroleum cracking. *OR*

(x) *Electronic applications*. Ferrimagnetic garnets of the type $3\text{Ln}_2\text{O}_3 \cdot 5\text{Fe}_2\text{O}_3$ are employed in microwave devices.

(b) Nuclear applications

Lanthanide elements and their compounds find many important nuclear uses, e.g., in nuclear fuel control, shielding and fluxing devices. Pr^{147} is used in the production of atomic battery.

Comparison between d- and f-block elements (particularly between 3d- and 4f-block elements)

Since both d- and f-block elements have partly filled inner orbitals, these have many properties in common, but there are significant differences also. Some of the differences are given below :

Actinides (5f-Block Elements)

DEFINITION

The elements in which the extra electron enters 5f-orbitals of $(n-2)$ th main shell are known as 5f-block elements, actinides or actinones. Thus strictly according to this definition of actinides only *thirteen* elements from Th_{90} ($5f^0 6d^2 7s^2$) to No_{102} ($5f^{14} 6d^0 7s^2$) should be the members of actinide series. However, all the *fifteen* elements from Ac_{89} ($5f^0 6d^1 7s^2$) to Lw_{103} ($5f^{14} 6d^1 7s^2$) are considered as the members of actinide series, since all these fifteen elements have similar physical and chemical properties. In fact the name actinides has been derived from actinium which is *prototype* of actinides as lanthanum is the prototype of lanthanides.

* POSITION OF ACTINIDES IN THE PERIODIC TABLE

The study of the position of the actinides in the periodic table can be made under the following two heads :

(i) Prior to the discovery of the trans-uranium elements (*i.e.* before 1940). The existence of the lanthanide series helped the discoverers to predict that another series of elements resulting from the addition of electrons to an $(n-2)$ f-shell (*i.e.* 5f-shell) should occur somewhere in the heavy elements region. Prior to the discovery of transuranium elements, the naturally occurring heaviest known elements namely Th_{90} , Pa_{91} , and U_{92} , were placed below Hf_{72} , Ta_{73} and W_{74} in IV B, V B and VI B groups of the periodic table, because these elements showed +4, +5 and +6 oxidation states and resembled Hf, Ta and W respectively in many of their properties. The then undiscovered transuranium elements with atomic numbers 93 to 100 were thus expected to occupy the positions in the periodic table below Re_{75} , Os_{76} , Ir_{77} , Pt_{78} , Au_{79} , Hg_{80} , Tl_{81} , and Pb_{82} respectively as shown in Fig. 10.1.

(ii) Following the discovery of the trans-uranium elements. The discovery of the element neptunium (Np_{93}) came in 1940 and this discovery was followed shortly by the discovery of plutonium (Pu_{94}) in 1941. The tracer chemical experiments with Np_{93} and Pu_{94}

IDENTIFICATION AND NUCLEAR SYNTHESIS OF TRANS-URANIUM ELEMENTS

Of all the naturally occurring elements, uranium has the highest atomic number equal to 92. For this reason this element continued to occupy the last position in the periodic table for a pretty long time. Since 1940 eleven elements with atomic numbers equal to 93, 94, 95, 96, 97, 98, 99, 100, 101, 102 and 103 respectively have been identified and synthesised by the transformation of naturally occurring elements by nuclear reactions. These man-made eleven elements are placed beyond uranium in the periodic table and are collectively called **trans-uranium elements**.

The year of identification of the first isotope, the names of the discoverers, the method of identification and synthesis of these trans-uranium elements are summarised in Table 10-1.

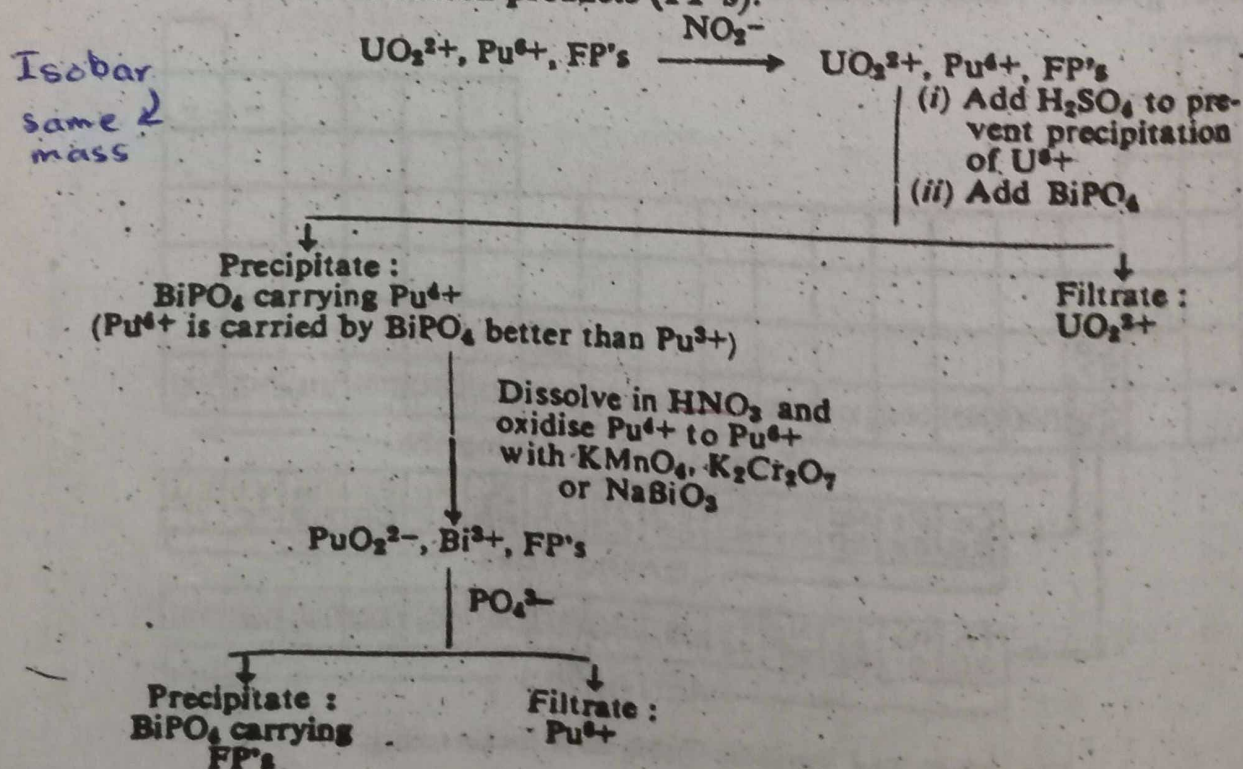
SEPARATION OF ACTINIDE ELEMENTS

Transuranium elements, produced as a result of nuclear reactions, are isolated from the target materials and irradiated nuclear fuels by the following methods:

(1) Precipitation method

Tri- and tetrapositive actinides can be precipitated as fluorides or phosphates from acidic solutions. Actinides in higher oxidation states either do not give a precipitate or form complexes. This method is particularly useful for the separation of the actinide elements of **U-Am** group. When the quantity of the actinide ion is insufficient to precipitate by itself, co-precipitation with a carrier like LaF_3 or BiPO_4 is adopted. The LaF_3 -coprecipitation method has been used for the separation of Np and Pu (obtained by neutron irradiation of uranium) from uranium and other fission products. The BiPO_4 -coprecipitation method has been discovered by Thomson and Seaborg and is used for the large scale preparation of Pu from U and fission products. This is shown in Flow-sheet 1.

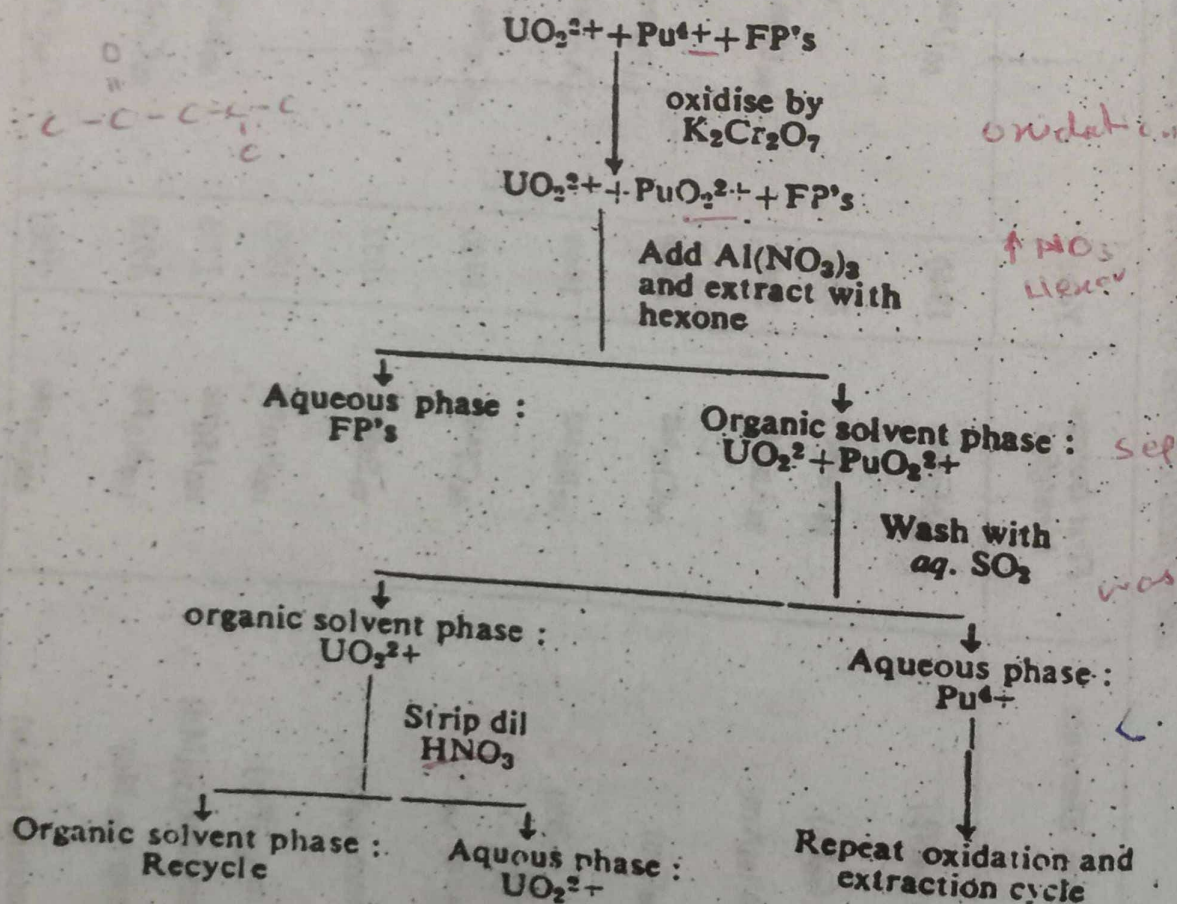
Flow-sheet 1. BiPO_4 -coprecipitation method for the separation of Pu from U and fission products (FP's).



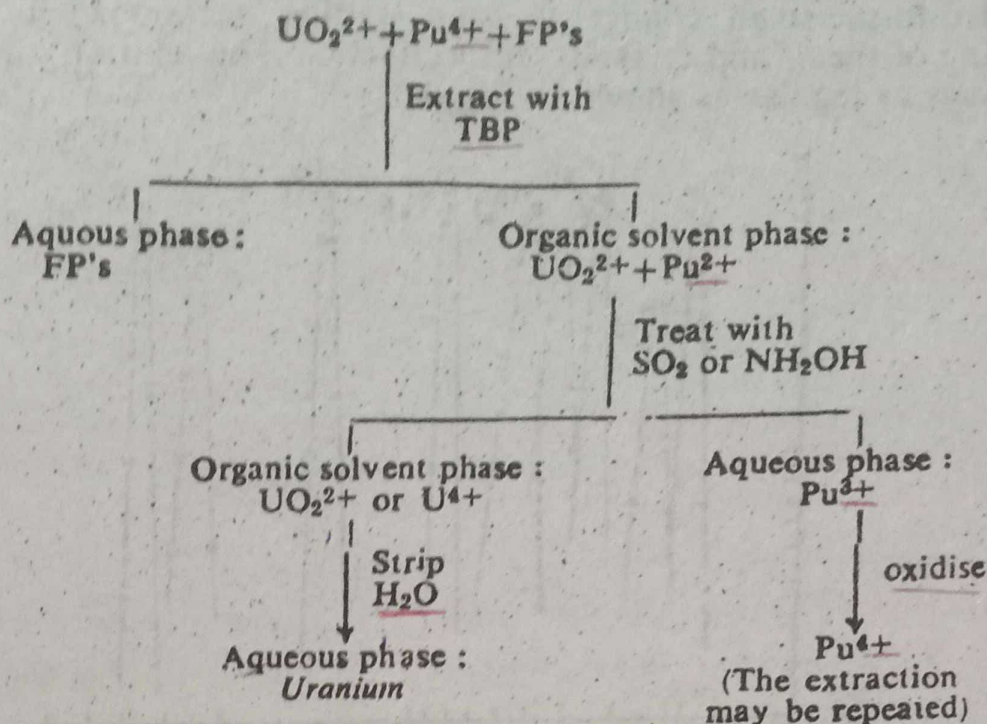
(2) Solvent extraction method

This method depends on the extractability of the various oxidation states of actinide elements. This technique finds extensive application in the recovery of U and Pu from used-up nuclear fuels. This process is based on the distribution of a metal between the aqueous solution and an organic solvent. Thus with methyl isobutyl ketone (hexone) Np^{3+} , Np^{5+} , Pu^{6+} and U^{6+} are extracted while Pu^{3+} is not extracted. Diethyl ether and tri-n-butyl phosphate (TBP) are other solvents which are used as extractants. Because of the high viscosity and density, TBP is used as 20% solution in kerosene. The method is preferentially applied to nitrate systems, because other ions like sulphate, perchlorate, fluoride etc. are strongly complexing and tend to retain the metal in aqueous solution. Hexone and diethyl ether require a high concentration of NO_3^- ions in the aqueous phase and it is achieved by adding $\text{Al}(\text{NO}_3)_3$ which has a high salt-out action. TBP is resistant to nitric acid oxidation and acts by itself as a salt agent. Solvent extraction of Pu and U by hexone and TBP is shown in Flow-sheets 2 and 3 respectively.

Flow sheet 2. Separation of Pu and U from fission products (FP's) by solvent extraction with hexane (Redox process).



Flow sheet 3. Separation of Pu and U from fission products (FP's) by solvent extraction with tributyl phosphate TBP (Purex process).



(3) Ion-exchange technique. Although this technique can be used to separate the actinide ions, this is best suited for separation of the trans-ameridium elements and also for some of the preceding elements, particularly for small amounts of material.

This method involves the following two steps:

(i) Lanthanide-actinide separation. The actinides as a group are separated from lanthanides by cation-exchange resin. Strong HCl is used as the eluting agent. The actinide ions form chloride complexes more easily and hence are eluted first.

Separation of actinides from lanthanides is now done on an anion-exchange resin with 10 M LiCl as eluent at elevated temperatures up to $\sim 90^\circ$. With the exceptions of Gd, Ho and of Cm, Es, the elution sequences proceed in the order of increasing atomic number. Thus La is absorbed least strongly.

(ii) Separation of individual actinide elements. The actinide ions, in general, can be separated from each other by removing from the cation exchange resin by elution with ammonium citrate, lactate, α -hydroxy isobutyrate and ethylene diamine tetracetate. If the activity is plotted against the number of drops of eluent, elution curves will be obtained as shown in Fig 10.5. Lw (atomic number, $Z=103$) is predicted to leave the column first, to be followed by No ($Z=102$), and so on down the scale of atomic numbers. Elution positions for Md ($Z=101$), Fm ($Z=100$), and down to Am ($Z=95$) are shown in the typical elution curves. Elution curves for lanthanides are also shown in the same figure for comparison. It may be noted that a very striking similarity occurs in the spacings of the corresponding elements in the two series (e.g. Am and Eu, Cm and Gd, Bk and Tb etc). It is this similarity that made possible for scientists to predict the elution positions of the elements from Bk to Md before their discovery, and which also makes possible today to predict the elution positions of elements with $Z=102$ and 103.

There is a distinct breakdown between Gd and Tb (lanthanide series) and between Cm and Bk (actinide series), which can be attributed to the small change in ionic radius occasioned by the half-filling of the 4f and 5f shells respectively. The elution order is not always as regular as shown in Fig. 10-5.

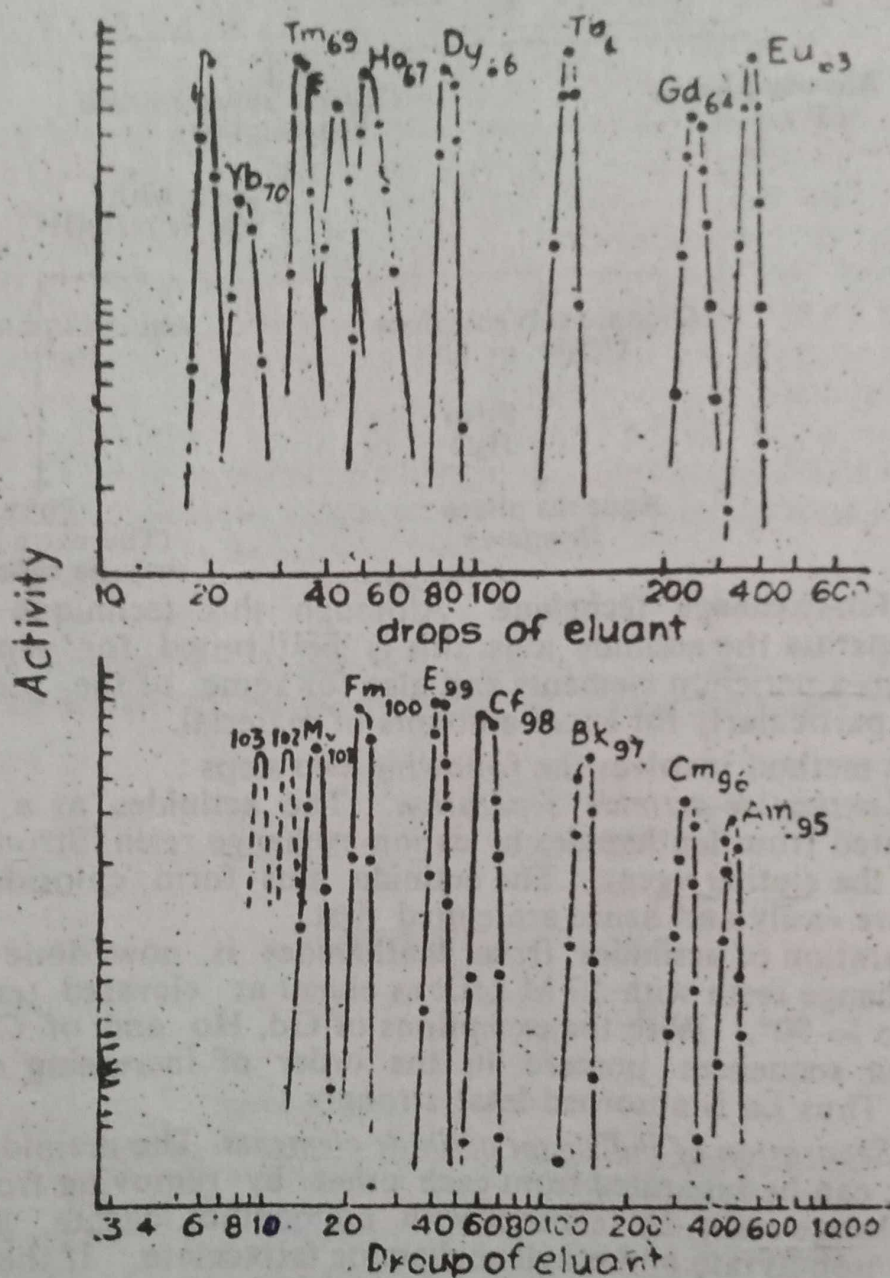


Fig. 10-5. Elution curves showing the elution positions of Ln^{3+} and An^{3+} ions eluted from Dowex-50 ion-exchange resin with ammonia alpha-hydroxy-isobutyrate. The dotted elution curves indicate the predicted elution positions of the then undiscovered elements with atomic numbers 102 and 103.

GENERAL PROPERTIES OF ACTINIDES AND THEIR COMPARISON WITH LANTHANIDES.

Some of the important properties of actinides are discussed below.

(1) **Electronic configuration and nature of bonding in actinide compounds.**

According to modern view 5f-orbitals in actinides start filling from protoactinium (Pa_{91}) and not from thorium (Th_{90}) as is evident

from the valence-shell electronic configurations of actinides given in Table 10-2. For the sake of comparison the valence-shell configurations of lanthanides are also given in the same table.

Table 10-2. *Electronic configurations of the neutral gaseous atoms of lanthanides and actinides.*

Lanthanides		Actinides		
Elements with atomic number	Valence shell configurations	Elements with atomic number	Complete configurations	Valence-shell configurations
La ₅₇	$4f^0 5d^1 6s^2$	Ac ₈₉	*[Rn]5f ⁰ 6d ¹ 7s ²	5f ⁰ 6d ¹ 7s ²
Ce ₅₈	$4f^2 5d^0 6s^2$	Th ₉₀	[Rn]5f ⁰ 6d ² 7s ²	5f ⁰ 6d ² 7s ²
Pr ₅₉	$4f^3 5d^0 6s^2$	Pa ₉₁	[Rn]5f ² 6d ¹ 7s ²	5f ² 6d ¹ 7s ²
Nd ₆₀	$4f^4 5d^0 6s^2$	U ₉₂	[Rn]5f ³ 6d ¹ 7s ²	5f ³ 6d ¹ 7s ²
Pm ₆₁	$4f^5 5d^0 6s^2$	Np ₉₃	[Rn]5f ⁴ 6d ¹ 7s ²	5f ⁴ 6d ¹ 7s ²
Sm ₆₂	$4f^6 5d^0 6s^2$	Pu ₉₄	[Rn]5f ⁶ 6d ⁰ 7s ²	5f ⁶ 6d ⁰ 7s ²
Eu ₆₃	$4f^7 5d^0 6s^2$	Am ₉₅	[Rn]5f ⁷ 6d ⁰ 7s ²	5f ⁷ 6d ⁰ 7s ²
Gd ₆₄	$4f^7 5d^1 6s^2$	Cm ₉₆	[Rn]5f ⁷ 6d ¹ 7s ²	5f ⁷ 6d ¹ 7s ²
Tb ₆₅	$4f^9 5d^0 6s^2$	Bk ₉₇	[Rn]5f ⁹ 6d ⁰ 7s ²	5f ⁹ 6d ⁰ 7s ²
Dy ₆₆	$4f^{10} 5d^0 6s^2$	Cf ₉₈	[Rn]5f ¹⁰ 6d ⁰ 7s ²	5f ¹⁰ 6d ⁰ 7s ²
Ho ₆₇	$4f^{11} 5d^0 6s^2$	Es ₉₉	[Rn]5f ¹¹ 6d ⁰ 7s ²	5f ¹¹ 6d ⁰ 7s ²
Er ₆₈	$4f^{12} 5d^0 6s^2$	Fm ₁₀₀	[Rn]5f ¹² 6d ⁰ 7s ²	5f ¹² 6d ⁰ 7s ²
Tm ₆₉	$4f^{13} 5d^0 6s^2$	Md ₁₀₁	[Rn]5f ¹³ 6d ⁰ 7s ²	5f ¹³ 6d ⁰ 7s ²
Y ₇₀	$4f^{14} 5d^0 6s^2$	No ₁₀₂	[Rn]5f ¹⁴ 6d ⁰ 7s ²	5f ¹⁴ 6d ⁰ 7s ²
Lu ₇₁	$4f^{14} 5d^1 6s^2$	Lw ₁₀₃	[Rn]5f ¹⁴ 6d ¹ 7s ²	5f ¹⁴ 6d ¹ 7s ²

*[Rn] indicates the complete configuration of Rn atom which is 2, 8, 18, 32, 18, 8.

Comparison of the electronic configurations of lanthanides and actinides shows both a substantial overall similarity and enhanced tendency for the lighter members namely Ac₈₉ to Np₉₃ to retain 6d electrons. The latter tendency results both from the even, smaller energy separation between the 5f and 6d orbitals in actinides than that between the 4f and 5d orbitals in lanthanides and from the difference in change in binding energy of the weakest bound (i.e. most readily removable) 6d and 5f electrons in actinides with atomic

number as shown in Fig. 10-6 in which a rough quantitative picture for the binding energies of the weakest bound 5f and 6d electrons in actinides is shown. Spectroscopic chemical and other data tell that 5f-level becomes progressively lower in energy compared to the 6d-level with increasing atomic number. The relative point of change is thus different in the two series and also the fall in energy and shrinkage in size are not so precipitous for 5f orbitals as for 4f-orbitals. The spatial extension of 5f orbitals relative to 6s and 6p orbitals is thus greater than that of 4f orbitals relative to 5s and 5p orbitals. The greater spatial extension of 5f orbitals has also been shown experimentally.

Since f shell is being filled in lanthanides (4f shell) and actinides (5f shell), the elements of these two series have close relationship in properties with one another. Thus Ac_{89} and La_{57} occupy corresponding positions as prototypes for the two series.

Although important similarities exist between the elements of the two series, very important differences also occur. These differences are due mainly to the relatively lower binding energies and less effective shielding of the 5f-electrons (in actinides) than those of 4f-electrons (in lanthanides).

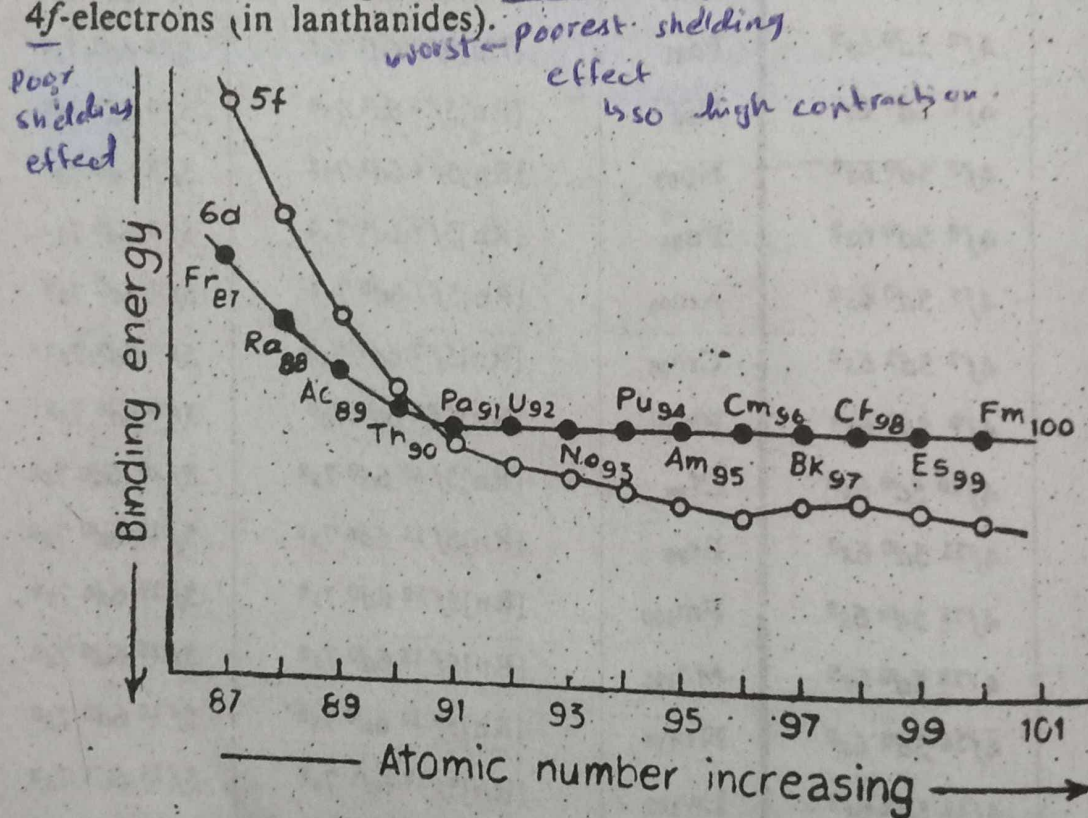


Fig. 10-6. Qualitative comparison of binding energies of 5f- and 6d-electrons. (Reproduced from Katz and Seaborg: "The Chemistry of the Actinide Elements," p. 465, Methuen, London, 1957).

The actinides have deeply seated incomplete 5f-orbitals compared to 4f-orbitals in lanthanides. 4f and 5f orbitals differ in their radial distribution functions. The radial distribution of 4f-orbitals shows a small shell of electron density below the main 5f-orbital (compare, for example, 1s and 2s radial functions). Therefore 5f-orbitals are more penetrating than 4f-orbitals. In lanthanides 4f-orbitals are well within the 6s and 5p orbital. Therefore there is no

question of an overlap between 4f and the surrounding ligands. The 5f-orbitals which are more penetrating than 4f-orbitals are little more diffuse at the periphery of the atom. Hence these orbitals are somewhat disturbed by ligands. It is well known that the energy difference between different levels and sub-levels decreases with increasing atomic number. Thus in the actinides the energies of 5f, 6d, 7s and 7p orbitals are but equal. This results in that unlike lanthanides, the bonding in the compounds of actinides can involve any or all of these orbitals. This fact is realised when we find that, while lanthanides virtually form no complexes or covalent compounds in which 4f-orbitals participate in bonding (lanthanides form mostly ionic compounds), the actinides form complexes not only with anions such as X^- , SO_4^{2-} etc., but also with π -bonding ligands such as alkyl phosphines, thioethers and cyclopentadiene by the use of 5f-orbitals.

(2) Oxidation States and Oxidation Potentials.

Observed oxidation states of actinide elements noted either in solution or in isolable compounds are summarised in Table 10.3.

Comparison of the oxidation states of lanthanides given in Table 9.4 p. 292 with those of actinides indicates that +3 oxidation state is the most common for both the series of elements. This oxidation state becomes increasingly more stable as the atomic number increases in the actinide series. The increasing stability of +3 oxidation state is illustrated by the increasing difficulty of oxidation above +3 oxidation state. This is clearly shown by the values of oxidation potentials (E° values) listed in the same Table. E° values have been recorded in 1M per chloric acid at 25°C. The standard electrode potentials for the lanthanides couple, Ln°/Ln^{3+} become steadily more positive with the increase of atomic number (due to lanthanide contraction), while for the actinides couple An°/An^{3+} these values become more positive from Ac to U and become less positive till Am. Am^{4+} is thus a more powerful oxidising agent than Ce^{4+} , when Pu^{4+} , Np^{4+} and U^{4+} are less powerful. U^{3+} is a strong reducing agent. SR

As regards lanthanides, no definite rule can be given about whether or not the +3 oxidation state of lanthanides increases in stability with the increase of atomic number.

Actinides show a greater multiplicity of oxidation states. Since in the first half of the actinide series (i.e. lower actinides), the energy required for the conversion $5f \rightarrow 6d$ is less than that required for the conversion $4f \rightarrow 5d$, the lower actinides should, on a comparative basis, show more higher oxidation states such as +4, +5, +6 and +7. Correspondingly, since in the second half of the actinide series (i.e. higher actinides), the energy required for the conversion $5f \rightarrow 6d$ is more than that required for the conversion $4f \rightarrow 5d$, the higher actinides should show more lower oxidation states such as +2. These expectations are in agreement with oxidation states given in Table 10.3. The second half of the series appears to parallel the lanthanides while the first halves of the two series are widely different from each other. more energy

Table 10-3. Various oxidation states shown by actinides and standard oxidation potentials, E° (in volts) for various couples in 1M per chloric acid at 25°C

Actinides with atomic numbers	Oxidation states (Less stable states are given in brackets)	E° values (in volts) for various couples (Estimated values are given in brackets)
Ac ₈₉	+3	$\text{Ac}^\circ \xrightarrow{2.6 \text{ volts}} \text{Ac}^{3+}$
Th ₉₀	+4	$\text{Th}^\circ \xrightarrow{1.90} \text{Th}^{4+}$
Pa ₉₁	(+4), +3	$\text{Pa}^\circ \xrightarrow{(0.9)} \text{Pa}^{4+} \xrightarrow{0.1} \text{PaO}_2^{2+}$ $\text{Pa}^\circ \xrightarrow{1.0} \text{PaO}_2^+$
U ₉₂	(+3), (+4), (+5), +6	$\text{U}^\circ \xrightarrow{1.80} \text{U}^{3+} \xrightarrow{0.631} \text{U}^{4+} \xrightarrow{-0.58} \text{UO}_2^{2+} \xrightarrow{-0.063} \text{UO}_2^{3+}$ $\text{U}^{4+} \xrightarrow{-0.32} \text{UO}_2^{2+}$
Np ₉₃	(+3), (+4), +5, (+6), (+7)	$\text{Np}^\circ \xrightarrow{1.83} \text{Np}^{3+} \xrightarrow{-0.155} \text{Np}^{4+} \xrightarrow{-0.739} \text{NpO}_2^+ \xrightarrow{-1.137} \text{NpO}_2^{2+}$ $\text{Np}^{3+} \xrightarrow{0.447} \text{NpO}_2^+$ $\text{Np}^{4+} \xrightarrow{-0.938} \text{NpO}_2^{2+}$
Pu ₉₄	(+3), +4, (+5), (+6), (+7)	$\text{Pu}^\circ \xrightarrow{2.03} \text{Pu}^{3+} \xrightarrow{-0.981} \text{Pu}^{4+} \xrightarrow{-1.172} \text{PuO}_2^+ \xrightarrow{-0.913} \text{PuO}_2^{2+}$

Actinides (5f-Block Elements)

Actinides (5f-Block Elements)

Am ₉₅	+2, (+3), (+4), (+5), (+6)	$\text{Am}^\circ \xrightarrow{(<2.7)} \text{Am}^{2+} \xrightarrow{>1.5} \text{Am}^{3+} \xrightarrow{-2.4} \text{Am}^{4+} \xrightarrow{-1.04} \text{AmO}_2^+ \xrightarrow{-1.60} \text{AmO}_2^{2+}$
Cm ₉₆	+3, (+4)	$\text{Cm}^\circ \xrightarrow{(>-2.80)} \text{Cm}^{4+}$
Bk ₉₇	+3, (+4)	$\text{Bk}^\circ \xrightarrow{(-1.6)} \text{Bk}^{4+}$
Cf ₉₈	+3	Black, main no 0.5
Es ₉₉	+3	has no stable nuclides
Fm ₁₀₀	+3	show with
Md ₁₀₁	+3	(0.2) $\text{Md}^{3+} \xrightarrow{(0.2)} \text{Md}^{4+}$
No ₁₀₂	+3	(-1.4) $\text{No}^{3+} \xrightarrow{(-1.4)} \text{No}^{4+}$
Lw ₁₀₃	+3	

The tripositive oxidation state occurs widely in each series. The two groups of elements, however, are not entirely comparable in this respect. The +3 state characteristic of lanthanides does not appear in aqueous solution of Th and Pa and this oxidation state becomes the predominantly stable oxidation state in aqueous solution of the actinides only when we reach Am. The elements from U to Am have several oxidation states, while there is no analogous example among lanthanides. These differences can readily be explained on the basis of the assumption that 7s-, 6d- and 5f-levels have almost the same energies in this region.

Since 5f-electrons are more easily removed than the 4f-electrons, for the actinide metals the lower oxidation states are less and the higher ions are more important compared to the lanthanide metals; e.g., even Cm gives Cm(IV) oxidation state in solids CmO_2 and CmF_4 , whereas only Gd(III) compounds are known. Cm^{4+} ion is also stable in solution.

Am +2 Oxidation state. Only americium (analogous to europium) is known to form a stable +2 state. This state is stable in CaF_2 only and has been studied by optical and electron spin resonance spectra. +2 state is uncommon for other actinides. An^{2+} ions resemble Ln^{2+} ions in their general chemistry.

+3 Oxidation state. +3 state is a general oxidation state for most of the actinides. For Th and Pa, +4 and +5 states respectively are important. An^{3+} ions resemble Ln^{3+} ions in their properties. A large number of isomorphous salts are given by the elements of both the series. Trichlorides and trifluorides of Ac, U, Np, Pu and Am are isomorphous. On hydrolysis all the halides give oxyhalides.

Ac, Pu and heavier elements give the oxides of An_2O_3 type which are isomorphous with Ln_2O_3 oxides.

Nitrates, perchlorates and sulphates are soluble while hydroxides, fluorides and carbonates are insoluble. Like lanthanides, actinides also form complexes with EDTA. Lanthanide-EDTA complexes are less stable than the corresponding actinide-EDTA complexes.

Th +4 Oxidation state. This is the principal oxidation state for Th and is a stable oxidation state upto Am. Am^{4+} and Cm^{4+} exist only as complexes in concentrated fluoride solution of low acidity. General chemistry of An^{4+} ions is similar to that of Ln^{4+} ions. The hydrated fluorides and phosphates of both An^{4+} and Ln^{4+} ions are insoluble. ThO_2 , PaO_2 , UO_2 , NpO_2 , AmO_2 , CmO_2 and BkO_2 have fluorite structure. The tetrachlorides and tetrabromides of Th, Pa, U and Np are only known, while tetraiodides of Th, Pa and U have been isolated. Oxyhalides of Th, U and Np can be prepared by heating AnX_4 with Sb_2O_3 . An^{4+} ions form complexes mostly with anionic ligands like HSO_4^- , NO_3^- , Cl^- etc.

+5 Oxidation state. This state is very important for Pa. Pa^{5+} resembles very much Na^{+} and Ta^{5+} . U, Np, Pu and Am also exist in +5 states, but these are less characterised. The only known pentahalides are those of Pa^{5+} and U^{5+} .

Fluoro anions of Pa, U, Np and Pu of the types AnF_6^- , AnF_8^{2-} and AnF_9^{3-} are known to exist in the solid state. Oxychlorides, AnOCl_3 ($\text{An} = \text{Pa, U and Np}$) are also known.

AnO_2^+ is the most important ion which contains An^{5+} cation. It has linear structure both in solid and solution. This monovalent dioxo cation forms many complexes with anionic and neutral ligands.

+6 Oxidation state. U, Np, Pu and Am show +6 oxidation state in divalent dioxo cation, AnO_2^{2+} . This cation is linear both in solid and solution. The simple molecular halide, UO_2F_2 , has the linear O—U—O group with fluorine bridges. The O—U bond distance is 1.75 to 2.00 Å. The overall structure is a flattened octahedron. Although AnO_2^{2+} cation is linear in shape, it forms complexes with exceptional geometries, e.g. four, five and six coordinated complexes are given by this cation.

+7 Oxidation states. +7 oxidation state is shown only by Np and Pu. Electrolysis or ozone oxidation of Np^{5+} or Np^{6+} in NaOH gives a green solution of NpO_6^{3-} which is slowly reduced to Np^{6+} at 25°C (E° for $\text{Np}^{7+}/\text{Np}^{6+} = 0.58$ volts in 1M NaOH). The existence of Np^{7+} ion has been confirmed by Mossbauer spectra.

Pu^{7+} ion is obtained by exposing a mixture of PuO_2 and Li_2O oxygen at 430°C when Li_2PuO_6 is formed. The chemistries of Np^{7+} and Pu^{7+} resemble those of Re^{7+} and Ie^{7+} .

(3) Atomic and Ionic Radii—Actinide Contraction

Crystal structure data have provided the basis for ionic radii of the actinide elements. In Table 10-4 atomic and ionic radii of the tripositive and tetrapositive actinide ions (i.e. An^{3+} and An^{4+} ions) are summarised. The atomic radii of lanthanide atoms and ionic radii of Ln^{3+} , Ln^{2+} and Ln^{4+} ions are also included in the same table for comparison. It may be noted that for An^{3+} and An^{4+} ions there is an actinide contraction (i.e. decrease in ionic radii of An^{3+} and An^{4+} ions), analogous to the lanthanide contraction, with increasing positive charge on the nucleus. This actinide contraction is due to the addition of successive electrons to an inner f-shell (i.e. 5f shell), so that the imperfect screening of the increasing nuclear charge by the additional 5f electrons results in a contraction of the outer or valence-orbital.

It may also be noted that the tripositive actinide ion (An^{3+} ion) is always somewhat larger than the formally analogous Ln^{3+} ion as shown in Fig. 10-7. The slight but readily detectable discontinuity at the Gd^{3+} ion does not have a parallel at the Cm^{3+} ion. Rather there is a continuing divergence in size among the An^{3+} ions as the nuclear charge increases. The ionic radii of Ln^{4+} and An^{4+} cations are shown together in Fig. 10-8.

The jumps in contraction between the consecutive elements in the actinides is greater than in the lanthanides. The greater contraction is due to the lesser shielding of 5f-electrons which are there-